



MODERN INSTITUTE OF TECHNOLOGY AND RESEARCH CENTRE, ALWAR

Name of faculty: Rekha Saini

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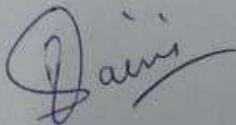
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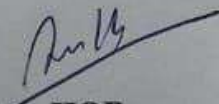
Topic covered: - Important Question Bank (NEW SCHEME PATTERN)

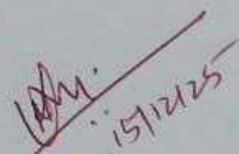
Section A: - 40 Important Questions

Section B: - 15 Important Questions

Section C: - 15 Important Questions


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B. Tech I SEM. IMPORTANT QUESTIONS (AS PER NEW PATTERN)

BRANCH: B. Tech (E-Batch)

SUBJECT: Managerial Economics & Financial Accounting (MEFA)

NAME OF FACULTY: Rekha Saini

SUB. CODE: 1FY3-05

SECTION-A

- Q.1 What is the Definition of Managerial Economics?
- Q.2 What is the meaning of demand?
- Q.3 What is Sole Proprietorship?
- Q.4 What is Limited Liability Partnership?
- Q.5 What do you mean by Joint Stock Company & Private Ltd. Company?
- Q.6 What is Earning per share?
- Q.7 What do you mean by Demand Forecasting?
- Q.8 What is the Laws of Variable Proportions
- Q.9 What is the State-owned enterprise?
- Q.10 What Is Financial Statement Analysis?
- Q.11 What is a Balance Sheet?
- Q.12 What is the Meaning and Definition of Elasticity of Demand ?
- Q.13 What is the Features of Double Entry?
- Q.14 What do you mean by Current Ratio?
- Q.15 What is net profit?

Q.16 What is the meaning of Monopolistic competition?

Q.17 What is the Sunk Costs?

Q.18 What is the meaning of Oligopoly?

Q.19 What is Quick Ratios?

Q.20 Describe the Barometric Method of demand forecasting.

Q.21 What is the Macro Economics?

Q.22 What is working capital?

Q.23 What is Supply?

Q.24 From the following information, prepare a Trading Account/ Profit & loss Account of M/s. ABC Traders for the Year ended 31st March, 2009:-

Opening Stock	Rs. 1,00,000
Purchases	6,72,000
Carriage Inwards	30,000
Wages	50,000
Sales	11,00,000
Returns inward	1,00,000
Returns outward	72,000
Closing stock	2,00,000
(Gross profit- 4,20,000)	

Q.25 What is the meaning of Complementary products ?

Q.26 What Is the Law of Supply?

Q.27 What is the meaning of production?

Q.28 What is the Accounting Costs?

Q.29 What is the meaning of Incremental Cost?

Q.30 What is gross profit?

Q.31 What is Pure or Perfect Oligopoly?

Q.32 Discuss the Delphi Method of demand forecasting.

Q.33 What do you mean by Organization and Industry Demand?

Q.34 Discuss about the Autonomous and Derived Demand.

Q.35 What do you mean by Substitute Product?

Q.36 What do you mean by Micro Economics?

Q.37 Discuss the features of Oligopoly Market.

Q.38 What is the difference between Monopoly and Oligopoly?

Q.39 What is the Meaning of Business Economics/ Managerial Economics?

Q.40 What is the Giffen Goods?



SECTION-B

- Q.1 What is the Definition of Managerial Economics?
- Q.2 Discuss about the Assumptions of Laws of Variable Proportions (Iso-quants and Iso-Cost Curve).
- Q.3 Explain the Different types of Concept of Cost in Economics.
- Q.4 Explain the Types/Kinds of Elasticity of demand.
- Q.5 Explain the features of Macro Economics.
- Q.6 What do you mean by Economic Problem and scarcity? Describe the Causes of Economic Problem in India.
- Q.7 Explain the Various Determinants Factor of Demand in the Market.
- Q.8 Explain the Limitations of Law of Demand.
- Q.9 A company is selling a pen. The company first determines the fixed costs (lease, property tax, and salaries) which sums up to ₹1,00,000. The variable cost determined by the company for one pen is ₹2 per unit. And, the pen is sold at a price of ₹10.
- Q.10 Explain the Features and Price determination under Perfect Market Competition.
- Q.11 Discuss the Key Concept of Law of Supply.
- Q.12 Explain the Different types of demand Schedule in the Market.
- Q.13 What Is the Law of Demand?
- Q.14 Explain the Cobb – Douglas and Leontief Production Functions.
- Q.15 Discuss about the Nature of Business Economics / Managerial Economics.

SECTION-C

- Q.1 Describe the Characteristics of Managerial Economics.
- Q.2 Explain the Features and Evaluation of Sole Proprietorship.
- Q.3 Describe the Features, Evaluation and types of Partnership.
- Q.4 Elaborate the Managerial economics and its relationship with other disciplines/ subjects.
- Q.5 Explain the Concept of Inflation. What is the difference between Inflation and Deflation.
- Q.6 Explain about the Joint Stock Company.
- Q.7 What is the Business Cycle? Discuss about the various Phases and features of Business Cycle.
- Q.8 Elaborate the Features and Advantages of Double Entry? What Is an Example of Double Entry?
- Q.9 What Is Ratio Analysis? Explain the Various types of Ratios with some Examples.
- Q.10 Discuss the Determinants and Method of demand forecasting?
- Q.11 Describe the Factors That Affect Supply and Exceptions to Law of Supply.
- Q.12 Explain the theory and factor of production and cost.
- Q.13 What is the Features of Monopoly Market. Describe the difference between Perfect, Monopoly and Monopolistic Competition?
- Q.14 From the following particulars, prepare a balance sheet of Mr. Somesh as at 31st March 2023: -

Capital ₹5,50,000 Drawings
₹10,000
Sundry Debtors ₹1,00,000

Sundry Creditors ₹80,000
Loan from Bank ₹20,000 Net
profit ₹1,60,000
Closing Stock ₹50,000
Plant and Machinery ₹1,50,000
Building ₹1,20,000
Land ₹ 3,00,000
Goodwill ₹ 50,000
Furniture and Fixtures ₹30,000

Q.15

Journalize the following transactions in the books of Laxman.

	₹
Business started with cash	80,000
Purchased furniture from Modern	
Furnishers for cash	10,000
Deposited in bank	30,000
Purchased goods for cash	5,000
Purchased goods on credit from Hari	4,500
Purchased goods from Suresh for cash	3,000
Purchased goods from Ajay	4,000
Sold goods for cash	7,000
Sold goods to Karim	6,500
Sold goods to Ganesh for cash	5,000
Sold goods to Bansi	3,500
Withdrew from bank for office use	5,000
Karim paid on account	5,000
Paid rent for residence	6,000
Paid to Ajay in settlement of his account	3,800

Answers
Section-A

Ans.1 What is the Definition of Managerial Economics: -

Managerial economics is defined as the branch of economics which deals with the application of various concepts, theories, methodologies of economics to solve practical problems in business management.

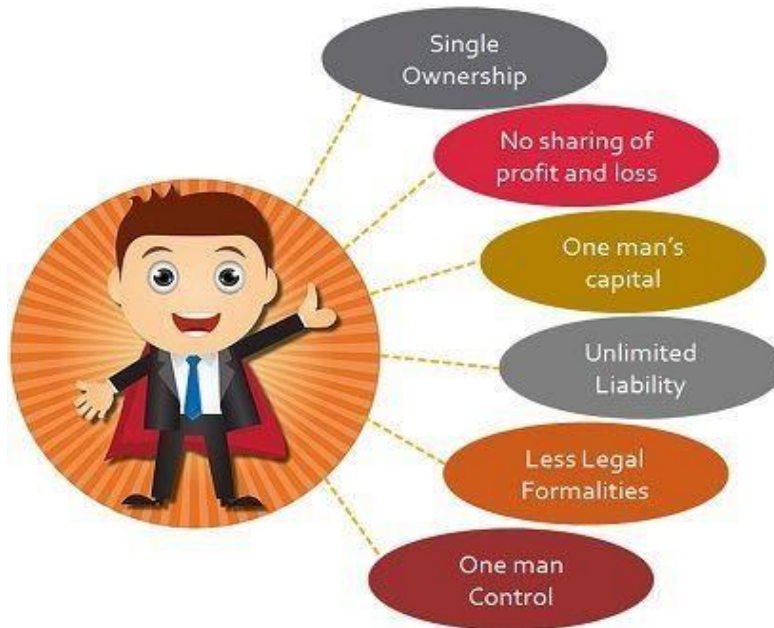
Ans.2 What is the meaning of demand?

It refers to the quantity of a good or service that consumers are willing and able to purchase at various prices during a given period of time.

- Desire of a commodity
- Ability to pay for it
- Willing to pay for it
- At the given price
- At a given time

Ans.3 What is Sole Proprietorship?

A Sole proprietorship is a business, owned, controlled and managed by a single individual. The owner has complete control over all aspects of the business and retains all profits. However, they also bear full responsibility for the business's liabilities and debts.



Ans.4 What is Limited Liability Partnership?

Limited liability partnerships (LLPs) are a common structure for professionals, such as accountants, lawyers, and architects. This arrangement limits partners' personal liability so that, for example, if one partner is sued for malpractice, the assets of other partners are not at risk.

Ans.5 What do you mean by Joint Stock Company & Private Ltd. Company?

Joint Stock Company: The simplest way to describe a joint stock company is that it is a business organisation that is owned jointly by all its shareholders. All the shareholders own a certain amount of stock in the company, which is represented by their shares.

Private Ltd. Company: According to Section 2(68) of the Indian Companies Act, 2013, a Private Company by its article is a company which:

1. Restricts the rights to transfer its shares.
 2. Has a minimum 2 and a maximum of 200 members, excluding the present and past employees.
 3. Prohibits any invitation to the public to subscribe for a company's securities.
- Examples of Private Limited Companies

The companies that run Flipkart, Ola, Snapdeal, Carat Lane, and Zoom Car are all private entities.

Ans.6 What is Earning per Share?

Earnings Per Share (EPS) means the profit earned by a company for each equity (ordinary) share. It shows how much profit belongs to each shareholder.

Formula of EPS

$$\text{EPS} = (\text{Net Profit after Tax} - \text{Preference Dividend}) / \text{Number of Equity Shares}$$

Example:

- Net profit after tax = ₹10,00,000
- Preference dividend = ₹1,00,000
- Number of equity shares = 1,00,000

$$\text{EPS} = (1000000 - 100000) / 100000$$

So, EPS = ₹9 per share.

Ans.7 What do you mean by Demand Forecasting?

Demand Forecasting refers to the process of predict the future demand for the firm's product.

The art of predicting the demand for a service or product in the future market is known as demand forecasting. This forecast is based on historical data, past customer behavior trends, and patterns of the present market. In short, it is the scientific estimation of future demand for products and not just merely guessing it.

Ans.8 What is the Laws of Variable Proportions: -

The Law of Variable Proportions is also known in the world of economics as the famous Law of Diminishing Returns. Many renowned economists and economic thinkers have opined their respective views on the concept of law of variable proportions.

Ans.9 What is the State-owned enterprise?

A state-owned enterprise or government-owned enterprise is a business enterprise where the government or state has significant control through full, majority, or significant minority ownership. Defining characteristics of SOEs are their distinct legal form and operation in commercial affairs and activities.

Ans.10 What is Financial Statement Analysis?

Financial statement analysis is the process of analyzing a company's financial statements for decision-making purposes. External stakeholders use it to understand the overall health of an organization and to evaluate financial performance and business value. Internal constituents use it as a monitoring tool for managing the finances.

Ans.11 What is a Balance Sheet?

The Balance Sheet is a statement that shows the financial position of the business. It records the assets and liabilities of the business at the end of the accounting period after the preparation of trading and profit and loss accounts.

Ans.12 What is the Meaning and Definition of Elasticity of Demand ?

The term elasticity is borrowed from physics. It shows the response of one variable with value to a change in other variables on which it is dependent. Elasticity is an index of reaction.

In economics the term elasticity refers to a percentage of the relative changes in two quantities. It measures the reaction of one variable to the changes in another variable

Ans.13 What is the Features of Double Entry?

- **Two Parties:** Two parties are involved, one is the receiver, and another is the giver. The receiving party is debited, and another party is credited. For example, A purchases goods from B, where A is a receiver party, and B is a giver party.
- **Equal Effect:** Each transaction should have an equal financial effect. The debit amount should be equal to the credit amount.
- **Separate Legal Entity:** This accounting system records the transaction separate from its owners.

- **Debit and Credit:** There are two aspects for recording any transaction, the Debit aspect, and the Credit aspect.

Ans.14 What do you mean by Current Ratio: -

The Current Ratio is one of the best-known measures of financial strength.

Current Ratio = Current Assets / Current Liabilities

Where,

Current Assets = Inventories + Sundry Debtors + Cash and Bank Balances + Bill Receivables + Accrued income + Loans and Advances / short term Investments + prepaid expenses

Current Liabilities = Creditors for goods and services + Short-term Loans + Bank Overdraft + Cash Credit + Outstanding Expenses + Provision for Taxation + Proposed Dividend + Unclaimed Dividend

Ans.15 What is net profit?

Net profit is the amount of money your business earns after deducting all operating, interest, and tax expenses over a given period of time. To arrive at this value, you need to know a company's gross profit. If the value of net profit is negative, then it is called net loss.

Ans.16 What is the meaning of Monopolistic competition?

Monopolistic competition is a situation in which the market, basically, is a competitive market but has some elements of a monopoly. In this form of market there are many firms that sell closely differentiated products. The examples of this form of market are Mobiles, Cosmetics, Detergents, toothpaste etc.

Ans.17 What is the Sunk Costs?

Sunk costs are the costs that cannot be recovered after sustaining. It includes the amount spent on conducting research and advertising.

Ans.18 What is the Meaning of Oligopoly: -

The term Oligopoly means 'Few Sellers'. An Oligopoly is an industry composed of only few firms, or a small number of large firms producing bulk of its output. Since, the industry comprises only a few firms, or a few large firms, any change in Price and Output by an individual firm is likely to influence the profits and output of the rival firms. Major Soft Drink firms, Airlines and Milk firms can be cited as an example of Oligopoly.

Ans.19 What is Quick Ratios:

The Quick Ratio is sometimes called the "acid-test" ratio and is one of the best measures of liquidity.

Quick Ratio or Acid Test Ratio = $\frac{\text{Quick Assets}}{\text{Current Liabilities}}$ Where,

Quick Assets = Current Assets - Inventories - prepaid Expenses

Current Liabilities = As mentioned under Current Ratio.

Ans.20 Describe the Barometric Method of demand forecasting.

This demand forecasting method uses historical sales data to predict future demand. It's based on the theory that demand for a product is influenced by previous demand for that product. To forecast demand using this method, you'll need sales data for a similar product from past years. You can then use this data to identify patterns and trends to help predict future demand. The barometric method is best suited for products with stable demand. If demand for your product fluctuates significantly from year to year, this method may not be as accurate.

Ans.21 What is the Macro Economics:-

Macro Economics is the study of the overall economic phenomena or the economy as a whole, rather than its individual parts. Accordingly, in Macro-Economics, we study the behavior of the large economic aggregates, such as, the overall levels of output, total consumption, total saving and total investment and also how these aggregates shift over time. It analyzes the overall economic environment in which the firms, governments and households make decisions. However, it should be kept in mind that this economic environment represents the overall effect of the innumerable decisions made by millions of different consumers and producers.

Ans.22 What is working capital?

Working Capital Ratio, also called the Current Ratio, measures a firm's short-term liquidity—its ability to pay short-term liabilities using short-term assets.

Working Capital Ratio = Current Assets ÷ Current Liabilities

Ans.23 What is Supply?

Supply is related to the price of the products, given that there is an incentive to supply at a higher price, as higher prices produce greater revenue and profits. Companies always want profit and, therefore, are more likely to manufacture more products at higher prices. When the price of goods or services is low, the supply is low, and when the price is high, the supply is also high.

Ans.24 From the following information, prepare a Trading Account/ Profit &

loss Account of M/s. ABC Traders for the Year ended 31st March, 2009:-

Opening Stock	Rs. 1,00,000
Purchases	6,72,000
Carriage Inwards	30,000



Wages	50,000
Sales	11,00,000
Returns inward	1,00,000
Returns outward	72,000
Closing stock	2,00,000
(Gross profit- 4,20,000)	

Ans:- Dr.

Cr.

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
Opening Stock	1,00,000	Sales 11,00,000 -Returns inward 1,00,000	10,00,000
Purchases 6,72,000 -Returns outward 72,000	6,00,000		
Carriage Inwards	30,000	Closing stock	2,00,000
Wages	50,000		
Total	7,80,000		12,00,000
Gross Profit b/f	4,20,000		

Ans.25 What is the meaning of Complementary products –

An increase in the price of one product will cause a decrease in the quantity demanded of a complementary product. Example: Increase in the price of bread will reduce the demand for butter. This arises because the products are complementary in nature.

Ans.26 What Is the Law of Supply?

The law of supply is a basic economic concept. It states that an increase in the price of goods or services results in an increase in their supply. Supply is defined as the quantity of goods or services that suppliers are willing and able to provide to customers. The law works like this: Rising prices mean that products become more profitable, assuming other factors such as production costs remain constant.

Ans.27 What is the meaning of production: -

Production is a very important economic activity. As we are aware, the survival of any firm in a competitive market depends upon its ability to produce goods and services at a competitive cost. One of the principal concerns of business managers is the achievement of optimum efficiency in production by minimizing the cost of production. The performance of an economy is judged by the level of its production. The amount of goods and services an economy is able to produce determines the richness or poverty of that economy. In fact, the standard of living of people depends on the volume and variety of goods and services produced in a country. Thus, the U.S.A. is a rich country just because its level of production is high.

Ans.28 What is the Accounting Costs?

Accounting costs are direct costs. They are also known as hard costs. The entrepreneur pays the cash directly for obtaining resources for production. It includes the cost of prices that are paid for the machines and raw materials, electricity bills, etc. These costs are treated as expenses.

Ans.29 What is the meaning of Incremental Cost

Incremental costs are the changes in future costs and that will occur as a result after a decision is made.

Ans.30 What is gross profit?

Gross profit is the profit a business makes after subtracting all the costs that are related to manufacturing and selling its products or services.

Gross profit = Total sales – Cost of Goods sold(COGS)

COGS= Opening stock +Purchases + Direct Expenses- Closing Stock

Ans.31 What is Pure or Perfect Oligopoly

Pure oligopoly is a form of market where few sellers sell a standardized product, and the actions of one firm directly affect the others.

Though, it is rare to find pure oligopoly situation, yet, cement, steel, Aluminum and chemicals producing industries approach pure oligopoly.

Ans.32 Discuss the Delphi Method of demand forecasting.

It is a demand forecasting technique that relies on the opinions of experts. The Delphi method is used to generate forecasts by asking a panel of experts their opinions on the future demand for a product. The experts' responses are then collated, and an average demand forecast is generated.

The Delphi method is ideal when there is little historical data available on which to base a demand forecast. The Delphi method can be used to generate forecasts for both short-term and long-term demand.

Ans.33 What do you mean by Organization and Industry Demand?

The demand for the product of an organization at given price above a point of time is known as organization demand. On behalf of example, the demand for Toyota cars is organization demand. The total of demand for products of all organizations in a particular industry is known as industry demand.

Ans.34 Discuss about the Autonomous and Derived Demand:

The demand of a product that is not associate with the demand of other products is known as autonomous or direct demand. The autonomous demand arises due to the natural desire of an individual to consume the product.

On behalf of example, the demand for food, shelter, clothes, and vehicles is autonomous as it increases due to biological, physical, and other personal needs of consumers. On the other hand, derived demand refers to the demand for a product that increases due to the demand for other products.

Ans.35 What do you mean by Substitute Product:-

– An increase in the price of one product will reason an increase in the demand for a substitute product. Example: Increase in price of tea will increase the demand for coffee and decrease the demand for tea.

Ans.36 What do you mean by Micro Economics?

Micro Economics is basically the study of the behavior of different individuals and organizations within an economic system. In other words, Microeconomics examines how the individual units make decisions as to how to efficiently allocate their scarce resources. Here, the focus is on a small number of or group of units rather than all the units combined, and therefore, it does not explain what is happening in the wider economic environment.

We mainly study the following in Micro-Economics:

- (i) Product pricing;
- (ii) Consumer behavior;
- (iii) Factor pricing;
- (iv) The economic conditions of a section of people;
- (v) Behavior of firms; and
- (vi) Location of industry.

Ans.37 Discuss the features of Oligopoly Market.

Features of Oligopoly: -

- 1. A Few Firms:-**Oligopoly as an industry is composed of few firms, or a few large firms controlling bulk of its output.
- 2. Firms are mutually dependent:-**Each firm in oligopoly market carefully considers how its actions will affect its rivals and how its rivals are likely to react. This makes the firms mutually dependent on each other for taking price and output decisions.
- 3. Barriers to the Entry of Firms:-**The main cause of a limited number of firms in oligopoly is the barriers to the entry of firms. One barrier is that a new firm may require huge capital to enter the industry. Patent rights are another barrier.

- 4. Non-Price Competition:-**When there are only a few firms, they are normally afraid of competing with each other by lowering the prices; it may start a Price War and the firm who starts the price war may ultimately lose. To avoid price war, the firm uses other ways of competition like: Customer Care, Advertising, Free Gifts etc. Such a competition is called non-price competition.

Ans.38 What is the difference between Monopoly and Oligopoly:-

BASIS FOR COMPARISON	MONOPOLY	OLIGOPOLY
Meaning	Monopoly is a form of market structure, where only one seller sells his distinctive product and dominates the entire market.	A market situation in which there are few firms in the market that sell either homogeneous or differentiated product and compete in the market.
Number of players	One	Two to ten
Product Differentiation	Extreme	None to substantial
Competition	Does not exist.	Little
Prices	High prices are charged.	Fair prices are charged.
Control over price	Very considerable	Some
Basis of setting price	Demand of consumers for the product.	Competitors prices.
Restrictions to entry	Due to economic, institutional, legal or any other reason.	Due to economies of scale.

Ans.39 What is the Meaning of Business Economics/ Managerial Economics:-

Business Economics is also referred to as managerial economics. It is application of economic theory & methodology. Every business involves decisions making as survival & success depends on sound decisions.

Ans.40 What is the Giffen Goods?

If the prices of basic goods like (sugar, wheat etc) on which the poor spend a large part of their income decline the poor increase the demand for superior goods. When the price of giffen goods fall its demand will also fall. There is a positive price effect in the case of giffen goods.

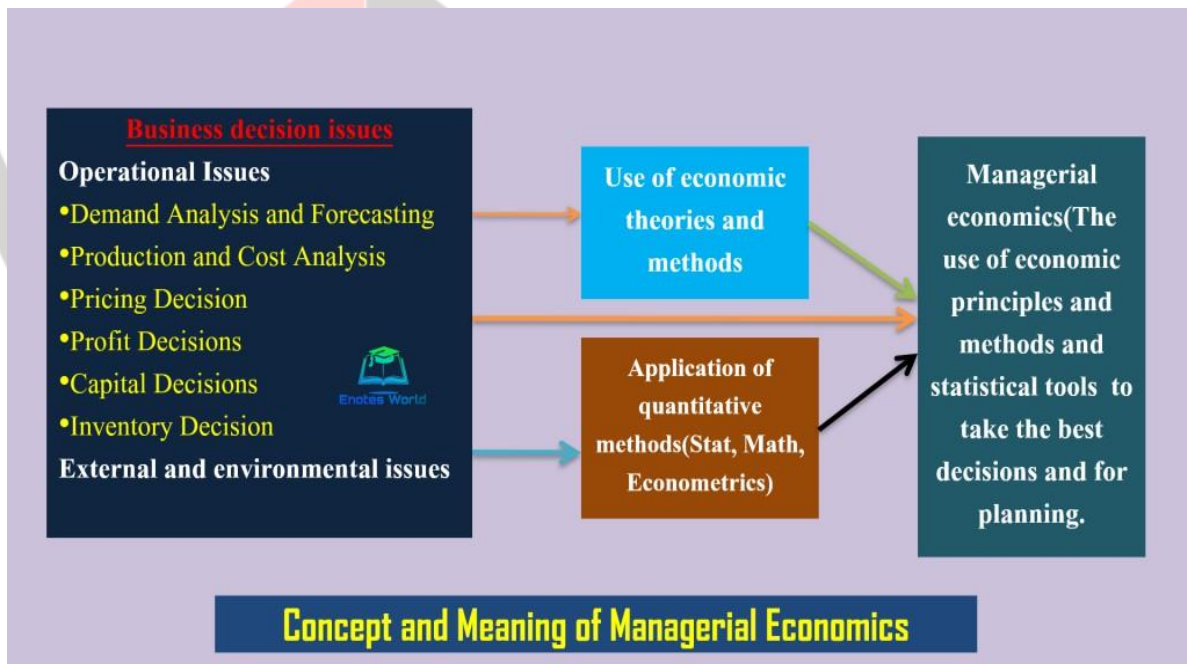


Answers Section-B

Ans.1 Explain the Concept of Managerial Economics.

Managerial Economics

- **Branch of Economics.** 
- 'Managerial Economics is the study of **Economic Theories, Principles and Concepts** which is used in **Managerial Decision Making.**'
- 'Managerial Economics is the **Application of various Theories, Concepts and Principles of Economics** in the **Business Decisions.**'
- It also Includes '**The Application of Mathematical and Statistical tools** in **Management decisions.**'



Ans.2 Discuss about the Assumptions of Laws of Variable Proportions (Iso-quants and Iso-Cost Curve)- :-

The law of variable proportions valid under certain assumptions.

(i) Constant State of Technology:

First, the state of technology is assumed to be constant. In the event of improvement in the technology, the marginal product may rise instead of diminishing.

(ii) Fixed Amount of Other Factors:

Secondly, there must be some inputs whose quantity should be kept fixed.

(iii) Varying the Factor Proportions:

Thirdly, the law is based upon the prospect of varying the proportions in which the various factors can be combined to produce a product.

Illustration of the Law:

The law of variable proportion is illustrated in the following table 3.1 and figure 3.1. Let us suppose that there amount of land is given in which more and more labour (variable factor) is used to produce rice.

Table 3.1: Law of Variable Proportion

Units of Labour	Total Product	Marginal Product	Average Product
1	2	2	2
2	6	4	3
3	12	6	4
4	16	4	4
5	18	2	3.6
6	18	0	3
7	14	-4	2
8	8	-6	1

It can be seen from the table 3.1 that up to the use of 3 units of labour, total

product increases at an increasing rate and beyond the third unit total product increases at a diminishing rate. This phenomenon is depicted by the marginal product which the addition is made to Total Product as a result of increasing the variable factor i.e. labour. Production Function with Two Variable Inputs:-

Isoquants: -

A production function with two variable inputs can be represented by a tool known as isoquants.

An Isoquant is a combination of two terms, namely, iso and quant. The meaning of 'Iso' is equal. The meaning of 'Quant' is quantity. Therefore, isoquant means equal quantity or equal product. Isoquants are defined as the curves which represent the different combination of inputs producing a particular quantity of output. Any point on the isoquant corresponds to the same level of output.

Assumptions of Isoquants:

1. There are only two inputs, labour and capital to produce a commodity.
2. The two inputs (L, K) can be substituted for one another at a diminishing rate, up to a certain limit.
3. Production function is continuous.

The isoquant illustrates all possible combinations of the two inputs (say labour and capital) capable of producing equal or a given level of output. Isoquants are also known as iso product curves or equal product curves or production indifferent curves. The following example in table 3.2 explains the concept of an isoquant.

Table 3.2

Combination Units of Capital and Total Output



Combinations	Units of labour	Units of capital	Total Output
A	20	1	100
B	15	2	100
C	11	3	100
D	8	4	100
E	6	5	100

In the above schedule of table 3.2, there are five possible combinations. All the five combinations yield the same level of output i.e. 100 units. 20 units of labour and 1 unit of capital produce 100 units. 15 units of labour and 2 units of capital also produce 100 units and so on. All the combinations are equally likely because all of them produce the same level of output i.e. 100 units.

Iso-Cost Curve: -

An iso-cost line reflects the various combinations of the two factors which the firm can buy at given prices with a given amount of money. Iso-cost line depicts all the combinations of labour and capital that the firm can purchase with a given outlay and at given prices. Thus iso-cost shows the prices of the two factors and the total amount of money to spend.

Let us suppose that a firm decides to spend Rs. 500 on 2 factors – capital and labour. If the weekly wage of a worker is Rs. 50, the firm can employ 10 workers. Similarly if one unit of capital costs Rs. 20, the firm can buy 25 units of capital. Thus the firm can spend the whole amount of Rs. 500 either on labour (10 workers) or on capital (25 units) or partly on labour and partly on capital. The iso-cost line is shown in the figure 3.7.

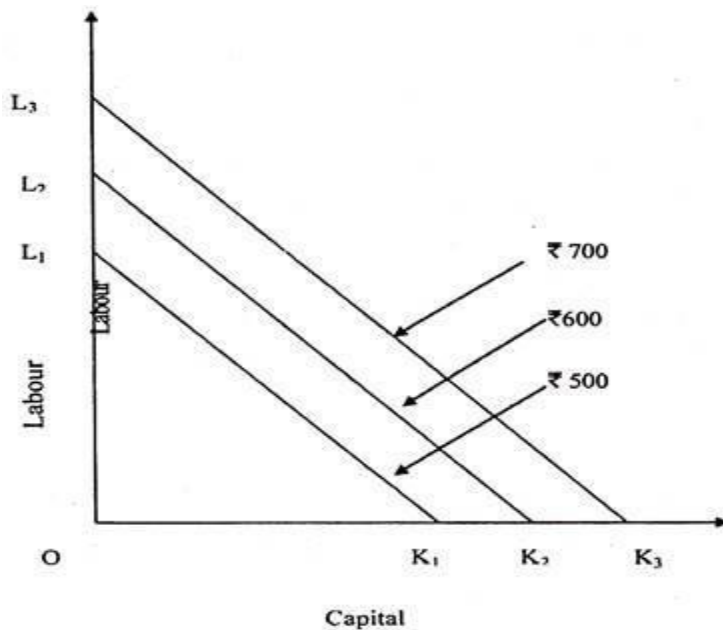


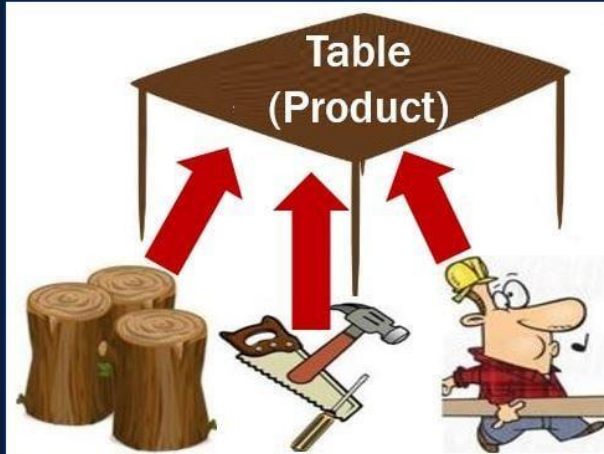
Figure 3.7 : The isocost line

The iso-cost line L_1, K_1 indicates an outlay of Rs.500. With Rs.500 the firm can buy either OL_1 units of labour or OK_1 units of capital or any combination of labour and capital between the extremes L_1 (K_1). Similarly the iso-cost line L_2, K_2 ... shows an outlay of Rs.600 which means that either 12 workers may be employed or 30 units of capital may be bought or some units of both capital and labour. Thus iso-cost line depicts all those possible combinations of capital and labour which the firm can use with the given amount of money.

Ans.3 Explain the Different types of Concept of Cost in Economics: -

The concept of cost is a key concept in Economics. It refers to the amount of payment made to acquire any goods and services. In a simpler way, the concept of cost is a financial valuation of resources, materials, risks, time and utilities consumed to purchase goods and services. From an economist's point of view, the cost of manufacturing any goods and services is often said to be the concept of opportunity cost.

What are Costs?



Resources used to make the product

Costs are the resources used to make a product. It is expressed in monetary terms.

Types of Cost Concept: -

The idea behind the concept of opportunity cost is that the cost of one item is the lost opportunity to do something else. For example, by being married to a person, one could lose the opportunity to marry some other person or by investing more capital in video games, one might lose the opportunity in watching movies.

Types of Cost Concept: -

The different types of cost concepts are:

1. Outlay costs and Opportunity costs
2. Accounting costs and Economic costs
3. Direct/Traceable costs and Indirect/Untraceable costs
4. Incremental costs and Sunk costs
5. Private costs and social costs

6. Fixed costs and Variable costs



Based on the Nature of Expenses

On the basis of nature, the following are the two types of cost:

- **Outlay Costs**

The authentic payments undergone by an entrepreneur in employing input are known as outlay costs. It includes costs on payments of fuel, rent, electricity, etc.

- **Concept of Opportunity Cost**

It is the value of the next best thing you give up whenever a decision is made by you.

Classification in Terms of Traceability

On the basis of traceability, the types of costs are:

1. **Direct Costs**

A direct cost is a cost that is related to the production method of a good or service. It is the opposite of an indirect cost.

2. **Indirect Costs**

Indirect costs are expenses that could not be traced back to a single cost object or cost source. They are also known as untraceable costs. However, they are extremely important as they affect the total profitability.

Concept of Costs in Terms of Treatment

1. **Accounting Costs**

Accounting costs are direct costs. They are also known as hard costs. The entrepreneur pays the cash directly for obtaining resources for production. It includes the cost of prices that are paid for the machines and raw materials, electricity bills, etc. These costs are treated as expenses.

2. **Economic Costs**

The economic cost is the combination of gains and losses of the

products. This cost is mainly used by economists to compare one with another.



Classification based on the Purpose

1. Incremental Cost

Incremental costs are the changes in future costs and that will occur as a result after a decision is made.

2. Sunk Costs

Sunk costs are the costs that cannot be recovered after sustaining. It includes the amount spent on conducting research and advertising.

Types of Cost Concept based on Players and Variability

1. Based on Payers

Private cost implies the cost that is sustained when an individual produces or consumes something. The business person spends his/her own private or business interests. The social cost is the cost to an entire society that results from a news event or a change in the policies.

2. In Terms of Variability

As the term predicts, fixed costs don't change in the volume of output. These costs are constant even with an increase or decrease in the volume of services/ goods produced or sold.

Ans.4 Explain the Types/Kinds of Elasticity of demand: -

3 Types of Elasticity of Demand: -

On the basis of different factors affecting the quantity demanded for a product, elasticity of demand is categorized into mainly three categories: **Price Elasticity of Demand (PED)**, **Cross Elasticity of Demand (XED)**, and **Income Elasticity of Demand (YED)**.

Let us look at them in detail and their examples.

1. Price Elasticity of Demand (PED):- Any change in the price of a commodity,

whether it's a decrease or increase, affects the quantity demanded for a product. For example, when there is a rise in the prices of ceiling fans, the quantity demanded goes down.



This measure of responsiveness of quantity demanded when there is a change in price is termed as the Price Elasticity of Demand (PED).

The mathematical formula given to calculate the Price Elasticity of Demand is:

$$\text{PED} = \% \text{ Change in Quantity Demanded} / \% \text{ Change in Price}$$

2. Income Elasticity of Demand (YED):- The income levels of consumers play an important role in the quantity demanded for a product. This can be understood by looking at the difference in goods sold in the rural markets versus the goods sold in metro cities.

The formula given to calculate the Income Elasticity of Demand is given as:

$$\text{YED} = \% \text{ Change in Quantity Demanded} / \% \text{ Change in Income}$$

The result obtained from this formula helps to determine whether a good is a necessity good or a luxury good.

3. Cross Elasticity of Demand (XED):- In a market where there is an oligopoly, multiple players compete. Thus, the quantity demanded for a product does not only depend on itself but rather, there is an effect even when prices of other goods change.

Cross Elasticity of Demand, also represented as XED, is an economic concept that measures the sensitiveness of quantity demanded of one good (X) when there is a change in the price of another good (Y), and that's why it is also referred to as Cross- Price Elasticity of Demand.

The formula given to calculate the Cross Elasticity of Demand is given as:

$$\text{XED} = (\% \text{ Change in Quantity Demanded for one good (X)}) / (\% \text{ Change in Price of another Good (Y)})$$

These three types of Elasticity of Demand measure the sensitivity of quantity demanded to a change in the price of the good, income of consumers buying the good, and the price of another good.



Ans.5 Explain the features of Macro Economics: -

Macro Economics is the study of the overall economic phenomena or the economy as a whole, rather than its individual parts. Accordingly, in Macro-Economics, we study the behavior of the large economic aggregates, such as, the overall levels of output, total consumption, total saving and total investment and also how these aggregates shift over time. It analyzes the overall economic environment in which the firms, governments and households make decisions. However, it should be kept in mind that this economic environment represents the overall effect of the innumerable decisions made by millions of different consumers and producers. A few areas that come under:-

Macro Economics are:-

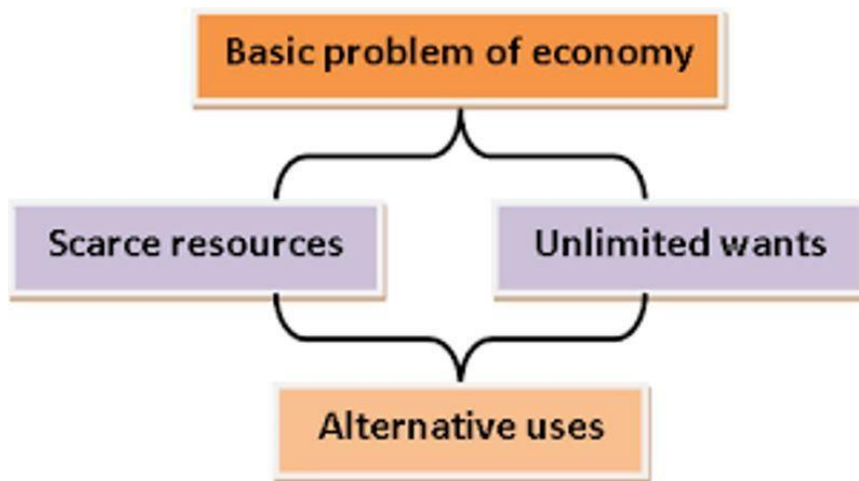
- (i) National Income and National Output;
- (ii) The general price level and interest rates;
- (iii) Balance of trade and balance of payments;
- (iv) External value of currency;
- (v) The overall level of savings and investment; and
- (vi) The level of employment and rate of economic growth.

While Business Economics is basically concerned with Micro Economics, Macro economic analysis also has got an important role to play. Macroeconomics analyzes the background of economic conditions in an economy which will immensely influence the individual firm's performance as well as its decisions. Business firms need a thorough understanding of the macroeconomic environment in which they have to function.

Ans.6 What do you mean by Economic Problem and scarcity? Describe the Causes of Economic Problem in India.

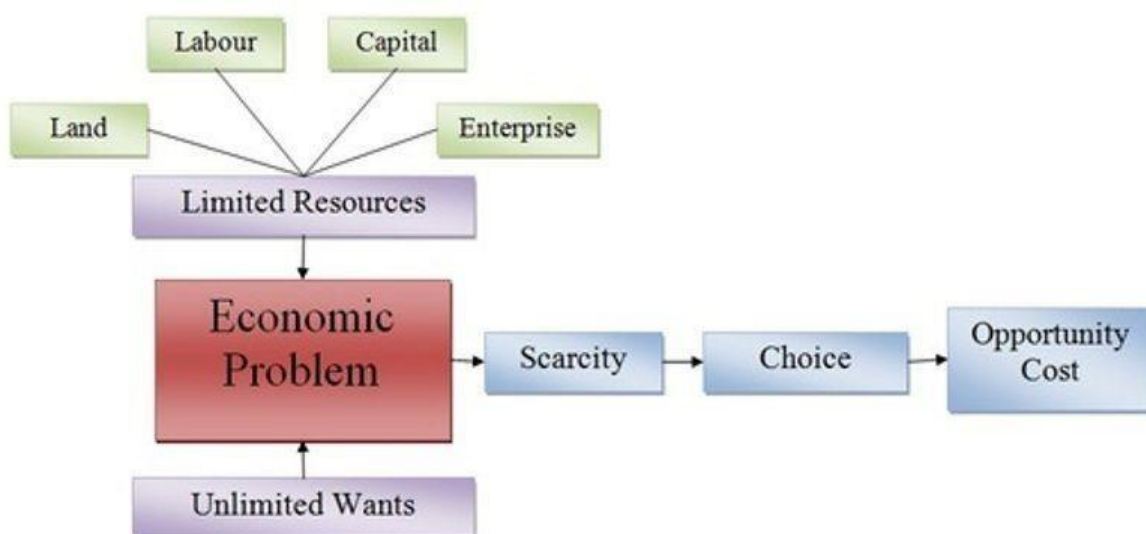
The Economic Problem: Scarcity and Choice: -





The chart shows the basic problem of economy:-

The basic problem of an economy deals with the needs and wants of a man being unlimited and the resources are scarce. The resources include the factors of production that are land, labour, capital and entrepreneurship.



Scarcity in Economics: -

The purpose of production is to satisfy one's want but as the resources are limited, not enough output is available to fulfil every man's want. This explains that human wants are unlimited which are not fulfilled by the limited resources as stated by the Law of scarcity.

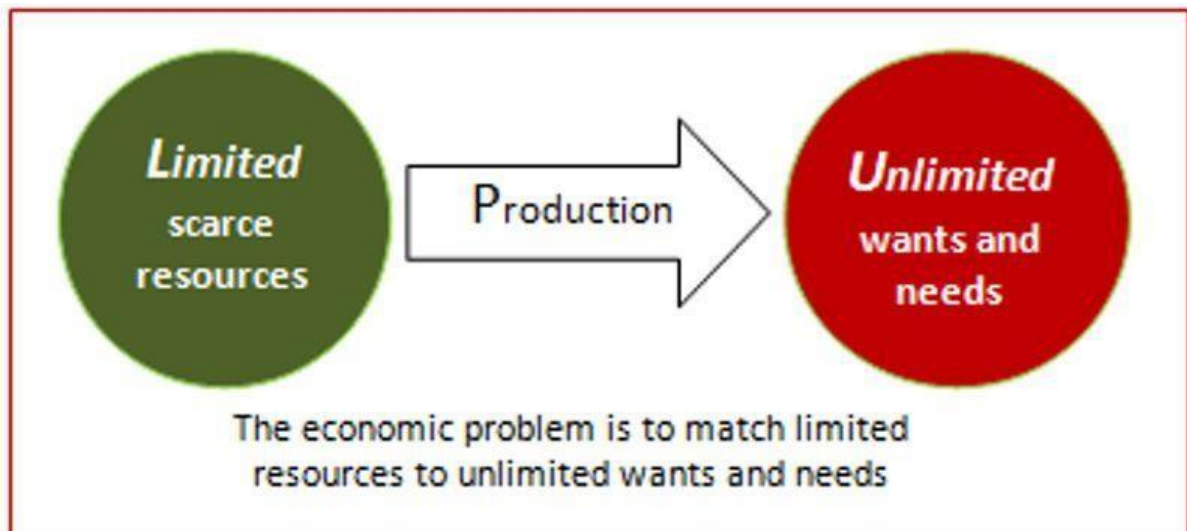


Chart:-This chart shows how law of scarcity arises

The demand is high as compared to the supply, and due to insufficient resources satisfaction is not achieved. To overcome this, the choice is made available to man to allocate their resources in such a way that maximum satisfaction can be achieved.

For instance, a man walks into a grocery store with ₹500, he would buy products in a way that when he walks out the products with him would be equal to the value of ₹500. He might want food grains, toiletries, milk, cooking essentials, etc but he would allocate the money available to him in such a way that he attains maximum satisfaction from his purchase.

Choices in Economics: -

Scarcity gives rise to the economic problem of choice. As there are limited resources, the choice is given to decide what one wishes to get by sacrificing one of its demand. When the choice is made there is sacrifice involved in it. The decision to consume a product also means a decision to not consume another. One product can only be consumed by giving up something in exchange. Opportunity Cost refers to the cost of sacrifice that is done to choose the next best alternative.

Causes of Economic Problem: -

- **Scarcity of resources:** Resources like labour, land, and capital are insufficient as compared to the demand. Therefore, the economy cannot provide everything that people want.



- **Unlimited Human Wants:** Human beings' demands and wants are unlimited which means they will never be satisfied. If a person's one want is satisfied, they will start having new desires. People's wants are unlimited and keep multiplying, therefore, cannot be satisfied because of limited resources.
- **Alternative Uses:** Resources being scarce, the same resources are used for different purposes. and it is therefore essential to make a choice among resources. For instance, petrol is used in vehicles and is also used for generators, running machines, etc. Therefore, the economy should now make a choice within the alternative uses.

List of Economic Problems:-

(A) What to produce?

- A country cannot produce all goods because it has limited resources.
- It has to make a choice between different goods and services.
- Every economy has to decide what goods and services should be produced.
- Example: If a farmer has a single piece of agricultural land, then he has to make a choice between two goods, i.e., whether to grow rice or wheat.
- Similarly, our government has to decide where to allocate funds, for the production of defence goods or consumer goods, and if both, then in what proportion.

(B) How to produce?

- This problem refers to the choice of technique of production. It arises when there is an availability of more than one way to produce goods and services.
 - There are mainly two techniques of production. These are:
 - Labour intensive technique(greater use of labour)
 - Capital intensive technique(greater use of machines)
 - Labour intensive technique promotes employment whereas capital intensive technique promotes efficiency and growth.
-

(C) For whom to produce?

- The society cannot satisfy all the wants of all the people. Therefore, it has to decide who should get how much of the total output of goods and services.



- Society has to make choice of whether luxury goods or normal goods have to be produced. This distribution or proportion directly relates to the purchasing power of the economy.

Ans.7 Explain the Various Determinants Factor of Demand in the Market: -

A determinant of Demand means factor's that effects the demand.

Demand for a commodity or service is decided by variety of things. All such factors are called as demand determinants. Price of the given commodity, prices of other substitutes and/or complements, future expected trend in prices etc.

- i. Price of the commodity
- ii. Price of related commodities
 - Complementary goods
 - Competing/ substitute goods
- iii. Income of the consumer
 - Normal goods
 - Inferior goods
- iv. Tastes & preference of buyers(Demonstration effect)
- v. Consumer's expectation
- vi. Other factors
 - Size of the population
 - Composition of population
 - Level of national income & its distribution
 - Consumer credit facility & interest rates

Thus, several factors are responsible for bringing changes in the demand for a product in the market. A corporate executive should have the knowledge and

knowledge about of these factors and forces so as to finalize his own production marketing and other business strategies.



1] **Price of the Product/Commodity:-** People use price as a parameter to form decisions if all other factors remain constant or equal. consistent with the law of demand, this suggests a rise in demand follows a discount in price and a decrease in demand follows a rise within the price of comparable goods.

2] **Income of the Consumers:-** Rising incomes cause an increase within the number of products demanded by consumers. Similarly, a drop by income is amid reduced consumption levels. The link between income and demand is not linear in character. Marginal utility determine the proportion of change in the demand levels.

3] Prices of related goods or services

- **Complementary products** – An increase in the price of one product will cause a decrease in the quantity demanded of a complementary product. Example: Increase in the price of bread will reduce the demand for butter. This arises because the products are complementary in nature.
- **Substitute Product** – An increase in the price of one product will reason an increase in the demand for a substitute product. Example: Increase in price of tea will increase the demand for coffee and decrease the demand for tea.

4] **Consumer Expectations:** -Expectations of a better income or expecting a rise in prices of products will cause a rise the number demanded. Similarly, expectations of a reduced income or a lowering in prices of goods will decrease the quantity demanded.

5] **Number of Buyers in the Market:** -The number of buyer has a major outcome on the total or net demand. As the number increases, the demand rises. Besides, this is true irrespective of changes in the price of commodities.

6] **Taste and preference:** -Taste and preference also affect demand of a commodity. Favorable taste results in increase in demand and unfavorable taste results in decrease in demand.

7] **Government policy:** -Government policy also affect demand of a commodity. If government imposes tax, then price of a product will increase and as a result demand will decrease.

8] **Distribution of income:** -If there is equal distribution of income then there will be more demand for a product and if there's unequal distribution of income, then there'll be less demand for a product.



Ans.8- Explain the Limitations of Law of Demand: -

When the prices of normal goods rise, the demand for them decrease, there are few cases where the law cannot operate. Following are the limitations of law of demand.

1. Prestige Goods

There are certain commodities like sports car or diamond, which are the sign of distinction and honor in any society. If the price of these goods increases the demand for them may be increase instead of falling.

2. Price Expectations

Expect a further increase in the price of a specific commodity they will go to buy more and more in spite of rising in price. In this case the violation of law is temporary. For Example Silver, Gold etc.

3. Ignorance of the Consumer

If the consumer is ignorant about the rise in price of goods, he may buy more of the commodity at higher price. For Examples:- Petrol, Diesel etc.

4. Giffen Goods

If the prices of basic goods like (sugar, wheat etc) on which the poor spend a large part of their income declining the poor increase the demand for superior goods. When the price of giffen goods fall it demand will also falls. There is a positive price effect in the case of giffen goods.

Ans.9 A company is selling a pen. The company first determines the fixed costs (lease, property tax, and salaries) which sums up to ₹1,00,000. The variable cost determined by the company for one pen is ₹2 per unit. And, the pen is sold at a price of ₹10.

Therefore, to determine the break-even point of Company, the pen will be:

Break-even point = Fixed cost/Price per cost – Variable cost

$$= ₹1,00,000 / (₹12 - ₹2)$$

$$= 1,00,000 / 10$$

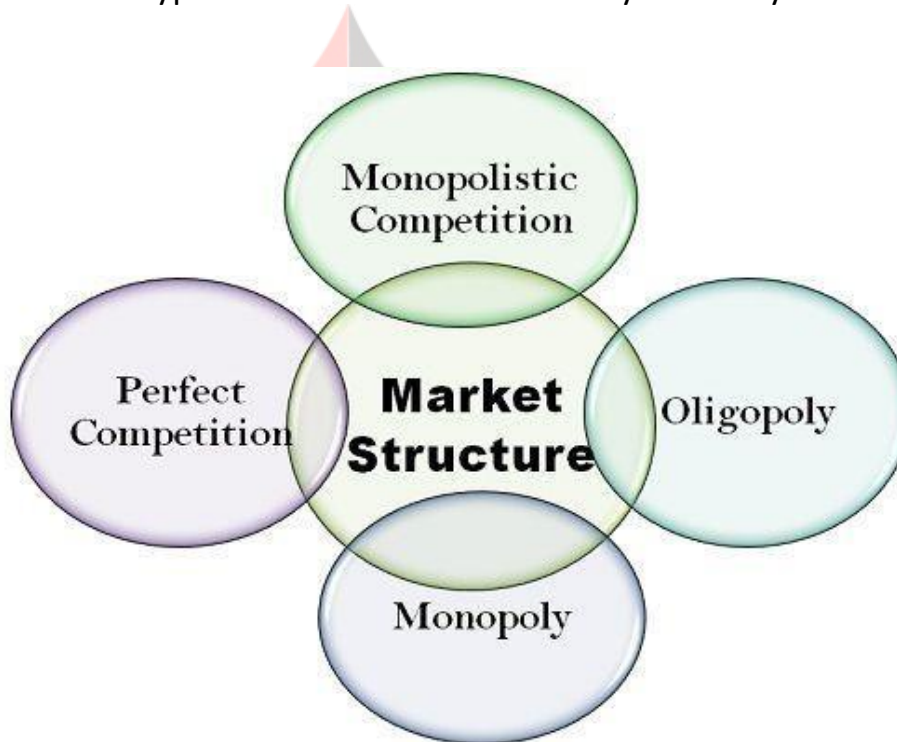
= 10,000

Therefore, with the given variable costs, fixed costs, and selling price of the pen, the company would need to sell 10,000 units of pens to break-even.

Ans.10 Explain the Features and Price determination under Perfect Market Competition: -

A variety of market structures will characterize an economy. Such market structures essentially refer to the degree of competition in a market.

There are other determinants of market structures such as the nature of the goods and products, the number of sellers, number of consumers, the nature of the product or service, economies of scale etc. We will discuss the four basic types of market structures in any economy.



Perfect Competition:-

Perfect Competition is a theoretical Market Structure that features unlimited contestability (No barriers to enter) and unlimited number of producers and consumers, and a Perfect Elastic Demand Curve. Perfect competition is a

market structure where an infinitely large number of buyers and sellers operate freely and sell a homogeneous commodity at a uniform price.



Features of Perfect Competition: -



1. **Many Buyers and Sellers** – There will always be a huge number of buyers and sellers in this form of marketplace. The advantage of having a large number of small-sized producers is that they cannot combine to influence the market price. If the quantity offered by an individual seller is very small compared to the total market produce, they cannot influence the market price independently.
2. **Homogeneity** – The product or service produced by the buyers in a perfectly competitive market should be homogenous in all respects. There should be no differentiation between them in terms of quantity, size, taste, etc., so that the products are perfect substitutes for each other. If a seller tries to charge a higher price for products that are so similar, they will lose their customers immediately.
3. **Free Entry and Exit** – Another condition of a perfectly competitive market is that no artificial restrictions prevent a firm's entry, or compel an existing firm to stay put when they want to leave. Their decision to enter, stay or leave the market depends purely on economic factors.
4. **Perfect Knowledge** – The buyers and sellers have perfect knowledge about the market conditions.

5. **Mobility of Factors of Production** – The factors of production like labour, raw materials and capital should have total mobility under perfect competition. The labour should have the freedom to move from one place (industry, market or production unit) to another depending on their remuneration.
6. **Transport Cost** – In the perfectly competitive market, the costs for transporting goods, services or factors of production from one place to another is either zero or constant for all sellers. The assumption is that all sellers are equally near or farther away from the market. Thus, the transport cost is uniform for all of them. The result is that the overall costs for production and the selling price are the same across the board.
7. **Absence of Artificial Restrictions** – There is no interference from the government or any other regulatory body to hinder the smooth functioning of the perfect competition. There are no controls or restrictions over the supply or pricing and the price can change solely based on the demand and supply conditions.
8. **Uniform Price** – There is a single uniform price for all products and services in a perfectly competitive market. The forces of demand and supply determine it.

Examples of Perfect Competition



- Agricultural markets
- Free software
- Street food vendors

Price Determination under Perfect Competition: -

Under perfect competition, many factors influence the determination of the price

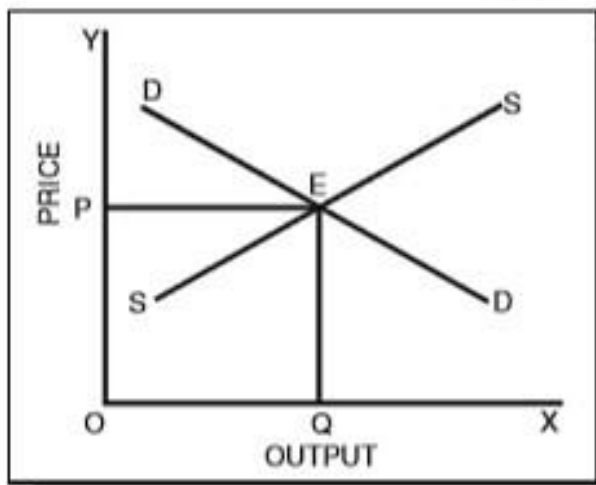
of goods. The equilibrium of the industry and the equilibrium of a firm as important factors behind price determination under perfect competition.

Equilibrium of the Industry under Perfect Competition: -

In economic terms, an industry consists of many independent firms. Each firm has a number of factories, farms or mines, as required. Each such firm in industry produces a homogeneous product. Equilibrium of the industry happens when the total output of the industry is equal to the total demand. In such a scenario, the prevailing price of a commodity is its equilibrium price.



We know that under competitive conditions, the interaction of demand and supply determines the equilibrium price as shown below:



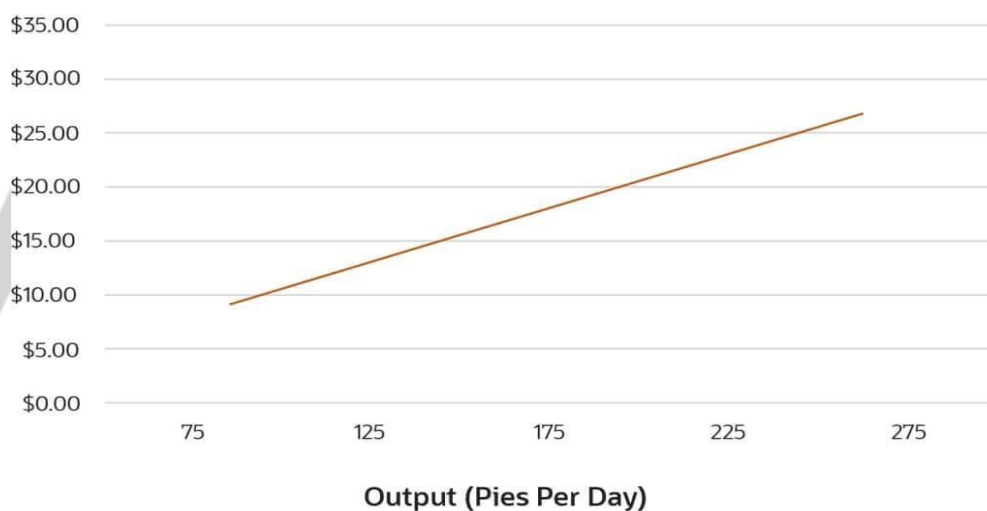
In Fig. 1 above, OP is the equilibrium price. Further, OQ is the equilibrium quantity sold at that price. Now, the equilibrium price is the price at which both the demand and supply are equal. In other words, no buyer, who wanted to buy at that price, goes dissatisfied and no seller, who wanted to sell his goods at that price, goes dissatisfied either.

Ans.11 Discuss the Key Concept of Law of Supply: -

- The law of supply states that an increase in the price of goods or services results in an increase in the quantity that suppliers make available to the market.
- Existing suppliers increase production of higher-priced goods to maximize profits, while new suppliers may also enter the market.
- The law of supply assumes that all other factors remain constant. In practice, many other factors can play into supply decisions, including rising production costs and market competition.
- Consider the example of a pizzeria that sells pasta dishes as well as pizzas. If

the price of pizza rises, and with it the profit per pie, the business may focus its resources on increasing the production of pizza — while decreasing the production of pasta offerings. As the price keeps rising, the pizzeria continues to increase the pizza supply because it can increase its profits by doing so. This relationship can be represented graphically as a supply curve, which shows the number of pizzas produced at different prices.

Pizza Supply Curve



As pizza prices rise, the pizzeria supplies more pies.

Ans.12 Explain the Different types of demand Schedule in the Market.

DEMAND SCHEDULE

Demand schedule is that the tabular statement shows various amount of a commodity which is demanded at different prices within a given period of your time.

Demand schedule is of 2 types

1. Individual demand schedule
2. Market demand schedule

INDIVIDUAL DEMAND SCHEDULE

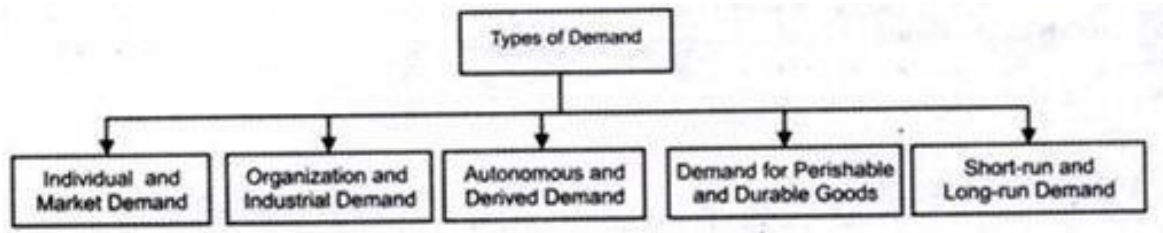
Individual demand schedule	
Price of ice-cream (₹)	Quantity demanded by 'A'(units)
1	4
2	3
3	2
4	1

Demand schedule is that the tabular statement show various amount of a commodity which is demanded at different prices within a given period of your time. Individual demand schedule may be a tabular statement showing the quantities of a given commodity which a consumer is willing to shop for in the least possible price within a given period of your time .

MARKET DEMAND SCHEDULE

Market demand schedule is a tabular representation showing the amount of a commodity demanded by all consumer in the market at all possible prices within a given period of time.

Market Demand Schedule				
Price of A (per unit in pkr)	Individual Demand (per day)			Market Demand (per day)
	X	Y	Z	
4	1	3	4	8
3	3	4	5	12
2	4	5	6	15
1	5	9	9	23

Types of Demand:-

The various types of demand are discussed as follows:

i. Individual and Market Demand:

Individual demand can be define as a quantity demanded by an person for a product at a particular price and within the specific period of time. For example, Mr. L demands 200 units of a product on Rs. 50 per unit in a week.

The individual demand of a product is partial by the price of a product, income of customers, and their tastes and preferences.

In simple language, market demand is the aggregate of individual demands of all the consumers of a product over a period of time at a specific price, while other factors are constant. on behalf of example, there are four consumers of oil (having a certain price). There are four consumers consume 30 liters, 40 liters, 50 liters, and 60 liters of oil respectively in a month. Hence, the market demand for oil is 180 liters in a month.

ii. Organization and Industry Demand:

The demand for the product of an organization at given price above a point of time is known as organization demand. On behalf of example, the demand for Toyota cars is organization demand. The total of demand for products of all organizations in a particular industry is known as industry demand.

For example, the demand for cars of various brands, like Toyota, Maruti Suzuki, Tata, and Hyundai, in India constitutes the industry' demand.

iii. Autonomous and Derived Demand:

The demand of a product that is not associate with the demand of other products is known as autonomous or direct demand. The autonomous demand

arises due to the natural desire of an individual to consume the product.

On behalf of example, the demand for petrol, diesel, and other lubricants depends on the demand of vehicles.

iv. Demand for Perishable and Durable Goods:

The goods are separated into two categories, perishable goods and durable goods. Perishable or non-durable goods are the products that have one use. For example, cement, coal, fuel, and eatables. On the opposite hand, durables are goods which will be used repeatedly.

On behalf of example, clothes, shoes, machines, and buildings. Perishable goods satisfy this demand of people. However, durables satisfy both present also as future demand of people. Therefore, consumers purchase durable items by considering its durability.

v. Short-term and Long-term Demand:

Short-term demand means to the demand for products that are used for a shorter duration of time or for current period. This demand depends on the present tastes and preferences of consumers.

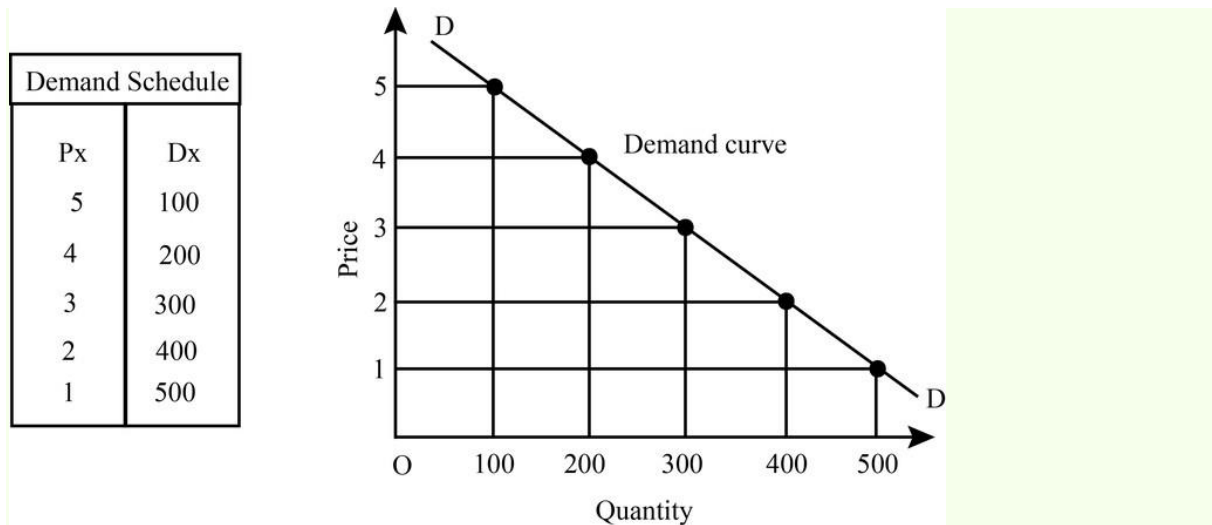
On behalf of example, demand for umbrellas, raincoats, sweaters, long boots is short term and seasonal in nature. On the opposite hand, long-term demand refers to the demand for products over an extended period of your time.

Ans.13 What Is the Law of Demand?

This law was first stated by Charles Davenant in 1699.

Law of demand states that there is an inverse relation between the price of a commodity and its quantity demanded, assuming all other factors affecting demand remain constant. It means that when the price of a good falls, the demand for the good rises and when price rises, the demand falls.

Law of demand may be explained with the help of the following demand schedule and demand curve: -



The above table and diagram show that as the price of the good reduces from Rs 5 to Rs 4, the demand for the good increases from 100 to 200 units.

Ans.14 Explain the Cobb – Douglas and Leontief Production Functions: -

The Cobb-Douglas production function is based on the empirical study of the American manufacturing industry made by Paul H. Douglas and C.W. Cobb. It is a linear homogeneous production function of degree one which takes into account two inputs, labour and capital, for the entire output of the manufacturing industry.

The Cobb-Douglas production function is expressed as:

$$Q = AL^a C^\beta$$

where Q is output and L and C are inputs of labour and capital respectively. A, a and β are positive parameters where $a > 0$, $\beta > 0$.

The equation tells that output depends directly on L and C, and that part of output which cannot be explained by L and C is explained by A which is the 'residual', often called technical change.

The production function solved by Cobb-Douglas had 1/4 contribution of capital to the increase in manufacturing industry and 3/4 of labour so that the

C-D(**Cobb- Douglas**) production function is

$$Q = AL^{3/4} C^{1/4}$$

which shows constant returns to scale because the total of the values of L and C is equal to one: $(3/4 + 1/4)$, i.e., $(\alpha + \beta = 1)$. The coefficient of labourer in the C-D function measures the percentage increase in (Q that would result from a 1 per cent increase in L, while holding C as constant.

Similarly, B is the percentage increase in Q that would result from a 1 per cent increase in C, while holding L as constant. The C-D production function showing constant returns to scale is depicted in Figure 20. Labour input is taken on the horizontal axis and capital on the vertical axis.

To produce 100 units of output, OC_1 units of capital and OL_1 units of labour are used. If the output were to be doubled to 200, the inputs of labour and capital would have to be doubled. OC_2 is exactly double of OC_1 and of OL_2 is double of OL_1 .

Criticisms of C-D(Cobb-Douglas) Production Function:

The C-D production function has been criticised by Arrow, Chenery, Minhas and Solow as discussed below:

1. The C-D production function considers only two inputs, labour and capital, and neglects some important inputs, like raw materials, which are used in production. It is, therefore, not possible to generalize this function to more than two inputs.
2. In the C-D production function, the problem of measurement of capital arises because it takes only the quantity of capital available for production. But the full use of the available capital can be made only in periods of full employment. This is unrealistic because no economy is always fully employed.

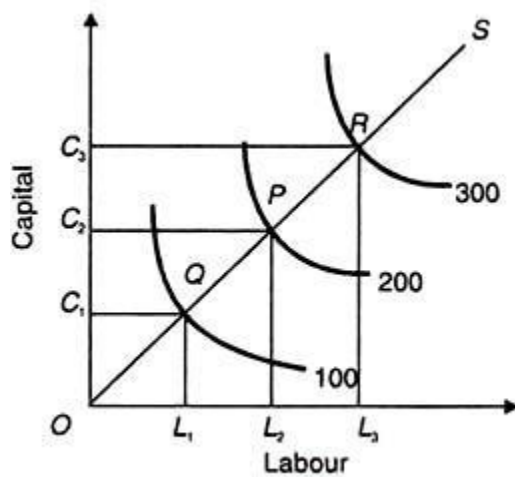


Fig. 20

3. The C-D production function is criticised because it shows constant returns to scale. But constant returns to scale are not an actuality, for either increasing or decreasing returns to scale are applicable to production.

It is not possible to change all inputs to bring a proportionate change in the outputs of all the industries.

4. The C-D production function is based on the assumption of substitutability of factors and neglects the complementarity of factors.

5. This function is based on the assumption of perfect competition in the factor market which is unrealistic. If, however, this assumption is dropped, the coefficients α and β do not represent factor shares.

6. One of the weaknesses of C-D function is the aggregation problem. This problem arises when this function is applied to every firm in an industry and to the entire industry. In this situation, there will be many production functions of low or high aggregation. Thus the C-D function does not measure what it aims at measuring.

Conclusion:

Thus the practicability of the C-D production function in the manufacturing industry is a doubtful proposition. This is not applicable to agriculture where for intensive cultivation, increasing the quantities of inputs will not raise output proportionately.

Leontief Production Function: -

Leontief production function uses fixed proportion of inputs having no substitutability between them. It is regarded as the limiting case for constant elasticity of substitution.

In economics, the **Leontief production function** or **fixed proportions production function** is a production function that implies the factors of production which will be used in fixed (technologically pre-determined) proportions, as there is no substitutability between factors. It was named after Wassily Leontief and represents a limiting case of the constant elasticity of substitution production function.

For the simple case of a good that is produced with two inputs, the function is of the form

$$q = \text{Min} \left(\frac{z_1}{a}, \frac{z_2}{b} \right)$$

where q is the quantity of output produced, z_1 and z_2 are the utilised quantities of input 1 and input 2 respectively, and a and b are technologically determined constants.

Example:-

Suppose that the intermediate goods "tires" and "steering wheels" are used in the production of automobiles (for simplicity of the example, to the exclusion of anything else). Then in the above formula q refers to the number of automobiles produced, z_1 refers to the number of tires used, and z_2 refers to the number of steering wheels used. Assuming each car is produced with 4 tires and 1 steering wheel, the Leontief production function is Number of cars = Min {1/4 times the number of tires, 1 time the number of steering wheels}.

Ans.15 Discuss about the Nature of Business Economics / Managerial Economics: -

The economic world is extremely complex as there is a lot of interdependence

among the decisions and activities of economic entities. Economic theories are hypothetical and simplistic in character as they are based on economic models built on simplifying assumptions. The following points will describe the nature of Business Economics

1. Business Economics is a Science:

Science is a systematized body of knowledge which establishes cause and effect relationships. Business Economics integrates the tools of decision sciences such as Mathematics, Statistics and Econometrics with Economic Theory to arrive at appropriate strategies for achieving the goals of the business enterprises. It follows scientific methods and empirically tests the validity of the results.

2. Based on Micro Economics:

Business Economics is based largely on Micro-Economics. A business manager is usually concerned about achievement of the predetermined objectives of his organization so as to ensure the long-term survival and profitable functioning of the organization.

3. Incorporates elements of Macro Analysis:

A business unit does not operate in a vacuum. It is affected by the external environment of the economy in which it operates such as, the general price level, income and employment levels in the economy and government policies with respect to taxation, interest rates, exchange rates, industries, prices, distribution, wages and regulation of monopolies.

4. Business Economics is an art

As it involves practical application of rules and principles for the attainment of set objectives.

5. Use of Theory of Markets and Private Enterprises:

Business Economics largely uses the theory of markets and private enterprise. It uses the theory of the firm and resource allocation in the backdrop of a private enterprise economy.

6. Pragmatic in Approach:

Micro-Economics is abstract and purely theoretical and analyses economic phenomena under unrealistic assumptions. In contrast, Business Economics is pragmatic in its approach as it tackles practical problems which the firms face in the real world.

7. Interdisciplinary in nature:

Business Economics is interdisciplinary in nature as it incorporates tools from other disciplines such as Mathematics, Operations Research, Management Theory, Accounting, marketing, Finance, Statistics and Econometrics.

8. Normative in Nature:

Economic theory has developed along two lines – positive and normative. A positive or pure science analyses cause and effect relationship between variables in an objective and scientific manner, but it does not involve any value judgment.

Section-C Answers

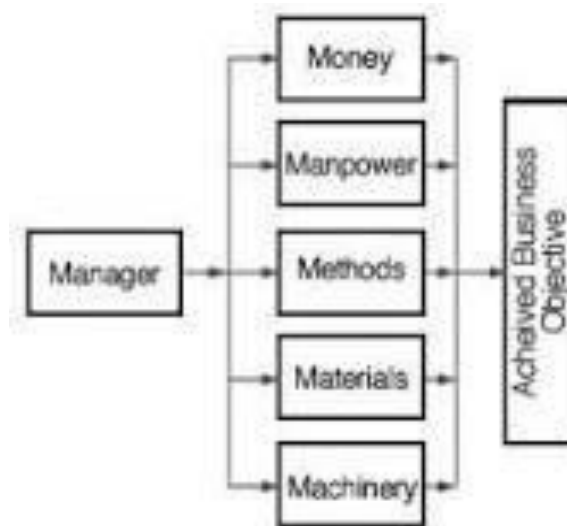
Ans.1 Describe the Characteristics of Managerial Economics: -

We will now look at the characteristics of managerial economics in brief.

1. **Art and Science:** -Managerial Economics requires a lot of creativity and logical thinking to come up with a solution. A managerial economist should possess the art of utilizing his capabilities, knowledge, and skills to achieve the organizational objective. Managerial Economics is also considered as a stream of science as it involves the application of different economic principles, techniques, and methods, to solve business problems.
2. **Microeconomics:** -In managerial economics, problems of a particular organization are looked upon rather than focusing on the whole economy. Therefore it is termed as a part of microeconomics.
3. **Uses Macroeconomics:** -Any organization operates in a market that is a part of the whole economy, so external environments affect the decisions within the organization. Managerial Economics uses the concepts of macroeconomics to solve problems. Managers analyze the macroeconomic factors like market conditions, economic reforms, government policies to understand their impact on the organization.
4. **Multi-disciplinary:** -Managerial Economics uses different tools and principles from different disciplines like accounting, finance, statistics, mathematics, production, operation research, human resource, marketing, etc. This helps in coming up with a perfect solution.
5. **Management oriented:** -Managerial economics is a tool in the hands of managers that aids them in finding appropriate solutions to business-related problems and uncertainties. As mentioned above, managerial

economics also helps in goal establishment, policy formation, and effective decision making. It is a practical approach to find solutions.





Five M's Showing Resources

Economics = “Balancing between unlimited wants & limited resources for the purpose of maximum satisfaction”

Economics is the study of the processes by which the relatively scarce resources are allocated to satisfy the competing unlimited wants of human beings in a society.

Every individual, every society and every country in this world faces the problem of choice this is because of two facts-

- (i) ‘Human beings have unlimited wants’; and
- (ii) ‘The means to satisfy these unlimited wants are relatively scarce’ form the subject matter of Economics.

Managerial Economics

Formula of Managerial economics = Management + Economics+ Decision Making

Here Decision making = Evaluation of available alternatives and selection of best alternative

Ans.2 Explain the Features and Evaluation of Sole Proprietorship

The main features of a Sole Proprietorship are as follows:

- **Legal Formalities** – No legal formalities are required to either commence or to shut down a sole proprietorship. But the owner must have a special license or certificate to run the business for specific occupations. For example, a sole proprietor planning to start a pharmacy must have a pharmacist's degree.
- **Unlimited Liability** – The sole proprietor is liable for the success or failure of their financial transactions. In case the proprietor takes a loan and fails to repay it, the creditors can attach the business owner's property to recover the loans.
- **Risk and reward** – A sole proprietor has complete ownership over the profits or losses from their firm's operations.
- **Control** – The rights and responsibilities of a sole proprietorship lies solely with its owner. No other person can interfere in the business activities of a sole proprietor without prior permission.
- **Continuity of business** – The existence of a business is related to its owner. Events like death, insolvency, imprisonment, terminal illness, etc., of the sole proprietor can adversely affect the business or force it to shut down permanently.

EVALUATION OF SOLE TRADER: -

Advantages and disadvantages of Sole Proprietorship

There are several benefits as well as limitations of running a sole proprietorship. We will discuss some of those points below:

Advantages –

- **Swift decisions** – A sole proprietor has complete responsibility in terms of making business decisions. It results in faster decision-making for the business as there is no need to consult multiple parties for every minor issue.

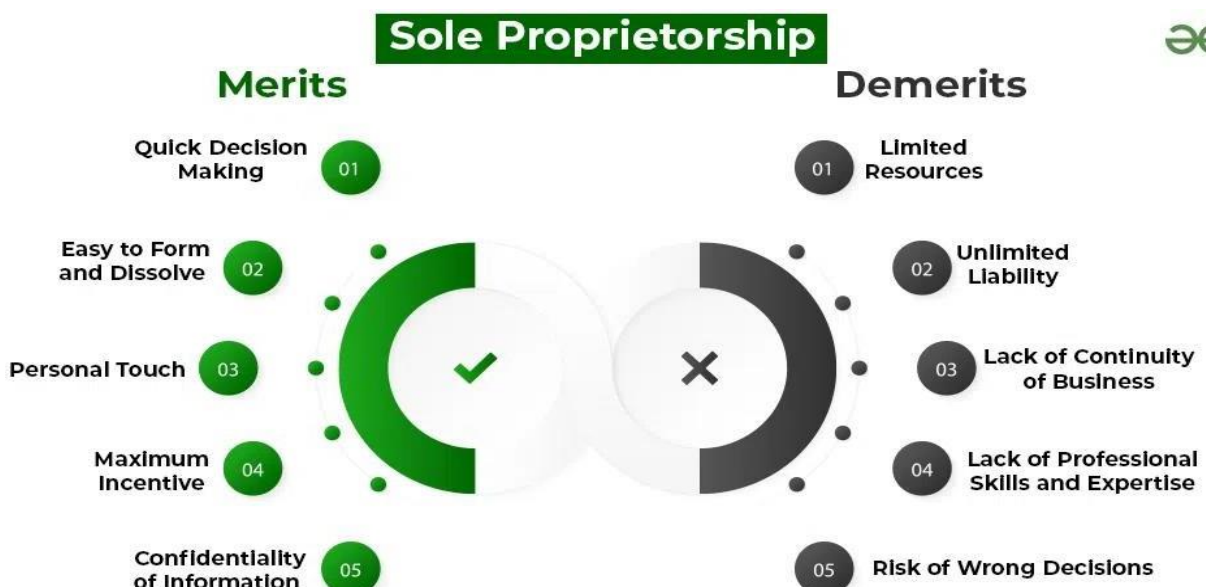
- **Confidentiality** – A sole proprietor can keep all business-related information to themselves as the business's only decision-maker. The law does not bind them to make the accounts of a sole proprietorship public.
- **Profit-sharing** – A sole proprietor has complete ownership of profits arising from business operations. They are not obligated to share profits with anyone else.



- **Fulfilment** – Since a sole proprietor is responsible for both risks and rewards of their business, even a minor success can give a greater feeling of pride and satisfaction than other business forms.

Disadvantages –

- **Lack of Resources** – It is challenging to raise vast amounts of capital in a sole proprietorship compared to a partnership or company. This form of business runs mainly on personal savings and borrowings made by its owner. Lack of adequate finances can become an obstacle in growing the business.
- **Dependence on owner** – The owner and their business are a singular entity in a sole proprietorship. In case of death, insolvency, imprisonment, etc., it can shut down if there is no successor or heir to continue the business.
- **Unlimited Liability** – If the proprietor cannot pay debts arising out of business from its assets, his/her personal property is also at stake. This results in sole traders taking zero or very minimal risks to ensure the survival of the business.
- **Management** – The proprietor has to perform most or all the activities related to the business-like purchase, client relationships, sales, marketing, accounting, etc.



Ans.3 Describe the Features, Evaluation and types of Partnership.**Meaning of Partnership: -**

In India, “**The Indian Partnership Act 1932**” regulates all aspects and activities of partnerships. This particular law indicates that a partnership is an association of two or more people or parties who have agreed to share the profits generated from the firm while managing it jointly or on behalf of other members.

Features of Partnership

The essential features and characteristics of a partnership are:

1. **Agreement:** The partnership arises out of an agreement between two or more persons.
2. **Profit sharing:** There should be an agreement among the partners to share the profits of the business.
3. **Lawful business:** The business to be carried on by a partnership must always be lawful.
4. **Membership:** There must be at least two persons to form a partnership. The maximum number is 20. But in case of banking business the maximum is 10 members.
5. **Unlimited liability:** The liability of every partner is unlimited, joint and several.
6. **Collective management:** The firm and the partners are one. When a contract is made in the name of the firm all the partners are responsible for it individually and collectively.
7. **non-transferability of shares:** A partner cannot transfer his share of interest to others without the consent of the other partners.

EVALUATION OF PARTNERSHIP: -**Advantages of Partnership**

The following are the advantages of partnership business:

1. **Easy to form:** A partnership firm can be formed without any legal formalities and expenses. Even if the firm is to be registered, the expenses are not much compared to company form of organization.
2. **Access to more capital:** A firm consists of more than one person. Therefore it can secure more capital from combined resources.
3. **Skill and talent:** Talented persons may be taken as partners. More skill and talent will be available..
4. **Division of labor:** Division of labor can be introduced which increases the efficiency in the management. One partner may take care of purchases, another sales, a third accounts and so on.
5. **Contact with customers:** All the partners in a firm may take part in the management of the business. So, they get in touch with the customers during the course of the business. It enables them to study the tastes and needs of the customers.
6. **Borrowing capacity:** The creditors will lend Loans not only on the basis of the firm's assets but also based on the personal properties of the partners. So the borrowing capacity of a firm is more.
7. **Incentive to work hard:** Every partner is liable for the debts of the firm. Also every partner has a share in the profits. This makes them to work hard for the success of the business.
8. **Expansion of business:** Due to the availability of sufficient finance and skill the business can be expanded very easily.
9. **Wise decisions:** In partnership, decisions are taken with the consultation of all the partners. So naturally the decisions are wiser and more beneficial.

10. **Co-operation between partners:** The partnership enables partners to provide mutual help to each other. Partners behave as members in a joint family.

11. **Flexibility:** Changes in the business can be adopted easily. There are no legal restrictions.

12. **Division of risks:** All losses and risks of the business are shared by all the partners. So risky ventures can also be taken up.

Disadvantages of Partnership

The following are the disadvantages of a partnership firm:

1. **Division of responsibility:** In a partnership the management is divided. As such responsibilities are also divided. Every partner might try to shift the burden on to the shoulders of others; finally none takes the responsibility properly.

2. **Delay in decisions:** Sometimes the partners may not agree with one another in taking decisions. As a result partners will not be in a position to take quick decisions.

3. **Lack of continuity:** A partnership gets dissolved on the death, insolvency, insanity or retirement of any partner. So, there is no guarantee for the continuity of the firm.

4. **No transferability of share:** In a firm the partner cannot transfer his share of interest to others without the consent of the other partners.

5. **Lack of secrecy:** It may not be possible to maintain secrecy in partnership because of the number of partners.

6. **Joint and several liability:** Every partner is jointly and separately liable for the firm's debts. In case of insolvency of partners, the solvent partners have to pay the debts of the insolvent partners also.

7. **Internal conflicts:** Differences and disputes among the partners are very common. These conflicts harm the firm as a whole.

8. **Misuse of assets:** The partners may use the assets of the firm for their personal purposes. Misuse of assets is harmful to business interests.

9. **Lack of public confidence:** A partnership firm is purely a private organization. It is not controlled or regulated by the Government. As such public may not have confidence in the firm.

Types of Partnerships:-

1. General Partnership

In a general partnership, all parties share legal and financial liability equally. The individuals are personally responsible for the debts the partnership takes on. Profits are also shared equally.

2. Limited Liability Partnership

Limited liability partnerships (LLPs) are a common structure for professionals, such as accountants, lawyers, and architects. This arrangement limits partners' personal liability so that, for example, if one partner is sued for malpractice, the assets of other partners are not at risk.

3. Limited Partnership

Limited partnerships are a hybrid of general partnerships and limited liability partnerships. At least one partner must be a general partner, with full personal liability for the partnership's debts. At least one other is a silent partner whose liability is limited to the amount invested. This silent partner generally does not participate in the management or day-to-day operation of the partnership.

Top examples of successful business partners: -

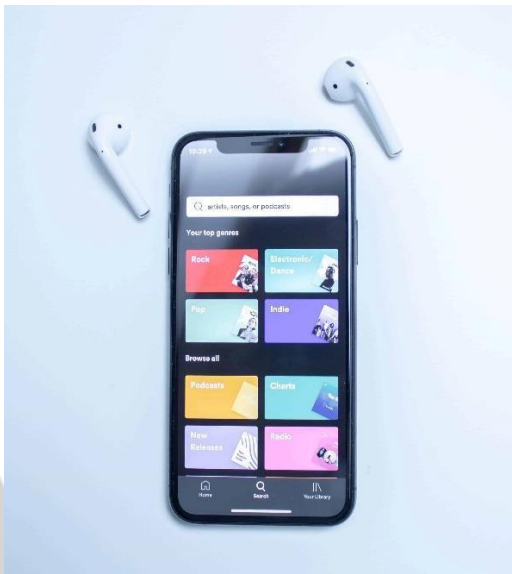
Red Bull & GoPro (2016)



Red Bull is an Austrian energy drink company founded in 1987. The CEO and the co-founder of the company are Dietrich Mateschitz.

He has a marketing degree, and before Red Bull, he worked for Unilever, Belindax and P&G. During his travels for Blandax, he discovered Krating Daeng, the drink that a few years later became Red Bull.

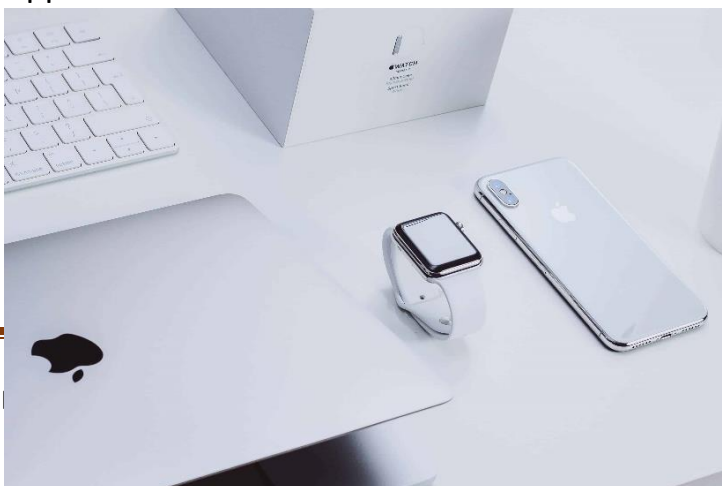
Starbucks & Spotify:-



Starbucks is an American multinational chain of coffeehouses and roastery reserves founded in 1971. In the early 1980s, Howard Schultz converted it from a coffee bean store into a coffee shop selling espresso-based drinks.

Today, the CEO is Kevin Johnson, a software engineer who worked for IBM, Microsoft and Juniper Networks.

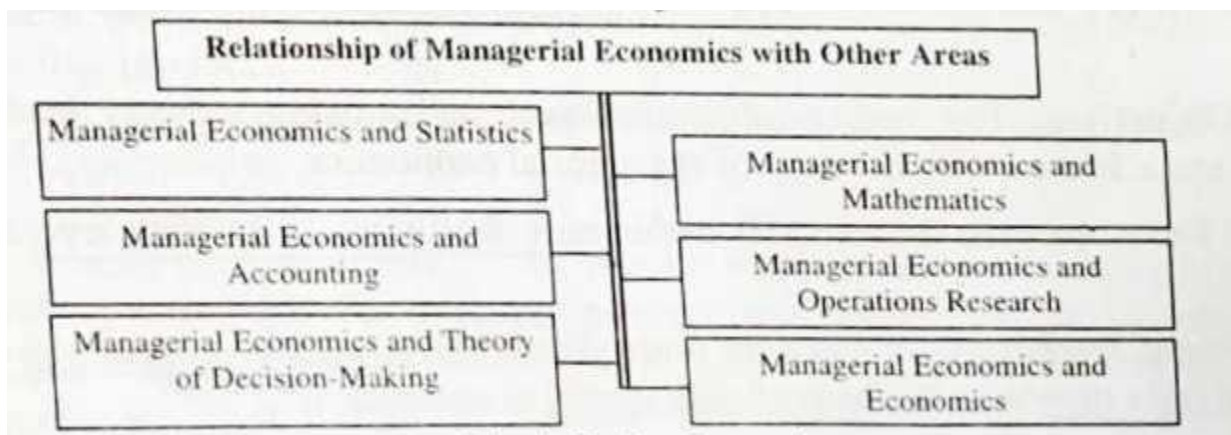
Apple & MasterCard:-



Apple is an American multinational technology company founded by Steve Jobs, Steve Wozniak and Ronald Wayne in 1976. Tim Cook has been the CEO of the company since 2011. Since then, he was able to double Apple's revenue and profits.

Ans.4 Elaborate the Managerial economics and its relationship with other disciplines/ subjects.

MANAGERIAL ECONOMICS AND ITS RELATIONSHIP WITH OTHER DISCIPLINES/ SUBJECTS: -



- 1. Managerial Economics and Economics:** -Managerial Economics is the economics applied to decision making to arrive at best solution. It may be studied as a special branch of economics, overcomes the gap between pure economic theory and managerial practice. There are two main branches of economics i.e. — micro- economics and macro-economics. Micro Economics is the study of individual units & also involves the study of decisions made in regard to the allocation of scarce resources and prices of various goods and services. The birth of managerial economics emerges from micro-economic theory. Whereas macro economics is the field that studies the behavior of economy as a whole.
- 2. Economics and Theory of Decision Making:** -Managerial economics uses both Economic Theory as well as Econometrics for rational decision making. In the course of business manager has to take various decisions such as related to production, labour, source of raw material, sources of capital, distribution channel, logistics etc. The decision-making process entails the evaluation of various possible alternatives & then selecting the best cause of action.

- 3. Managerial Economics and Operations Research:-**Operations research involves the application of mathematical techniques in solving business problems. Operations Research enables construction of theoretical-models that facilitate the decision-making process. Although the roots of operations research lie in studies of military ,it is now mostly used in planning and controlling various business activities. Various techniques of operations research which are found to be useful are linear programming, game theory, dynamic programming, input-output analysis, simulation techniques.
- 4. Managerial Economics and Statistics:-**Managerial economics is closely related with statistics. In business Managers have to go with the estimates of demand, market trends, probabilities regarding supply. Statistics as an important tool helps to study the various functional relationships and also facilitate forecasting. In this regard Managerial Economics make use of various statistics tools like Trend projection, correlation, multiple regression, measures of central tendency like the mean, median, mode, measures of dispersion to arrive at the best solution to the business problems.
- 5. Managerial Economics and Accounting:-**Managerial economics and accounting goes hand in hand. While continuing with the business a manager has to undergo various transactions of financial nature. The recording of these transactions is facilitated by the subject i.e. Accounting. Accounting records all transactions of financial nature on routine basis to provide the realistic view of the business financial position and thus enables decision making through the use of accounting data. The accounting techniques are very essential for the success of the firm because profit maximisation is the major objective of the firm.
- 6. Managerial Economics and Mathematics:-**Mathematics is yet another important subject closely related to managerial economics. The important concepts of mathematics used by a managerial economist are geometry, algebra and calculus, logarithms and exponential, vectors and determinants, input-output tables. Managerial economics make use of various techniques of mathematics for the derivation and exposition of economic analysis, estimation and prediction of economic factors .all the techniques of mathematics provides the logical and reliable results.

Ans.5 Explain the Concept of Inflation. What is the difference between Inflation and Deflation.

CONCEPT OF INFLATION: -**Definition**

- According to classical writers inflation is a situation when too much money chases too few goods.
- It is an imbalance between money supply and Gross Domestic Product.
- As per Keynes inflation is an imbalance between aggregate demand and aggregate supply.
- In an economy, if the aggregate demand For goods and services exceeds aggregate supply, then Prices will go on rising.

What is Inflation and How it Works?

Inflation is an economic indicator that indicates the rate of rising prices of goods and services in the economy. Ultimately it shows the decrease in the buying power of the rupee. It is measured as a percentage.

Types of Inflation: -**What are the 3 Types of Inflation?**

The three types of Inflation are Demand-Pull, Cost-Push and Built-in inflation.

1. **Demand-pull Inflation:** It occurs when the demand for goods or services is higher when compared to the production capacity. The difference between demand and supply (shortage) result in price appreciation.
2. **Cost-push Inflation:** It occurs when the cost of production increases. Increase in prices of the inputs (labour, raw materials, etc.) increases the price of the product.
3. **Built-in Inflation:** Expectation of future inflations results in Built-in Inflation. A rise in prices results in higher wages to afford the increased cost of living. Therefore, high wages result in increased cost of production, which in turn has an impact on product pricing. The circle hence continues.

What are the Main Causes of Inflation?

1. **Monetary Policy:** It determines the supply of currency in the market. Excess supply of money leads to inflation. Hence decreasing the value of

the currency.

2. **Fiscal Policy:** It monitors the borrowing and spending of the economy. Higher borrowings (debt), result in increased taxes and additional currency printing to repay the debt.
3. **Demand-pull Inflation:** Increases in prices due to the gap between the demand (higher) and supply (lower).
4. **Cost-push Inflation:** Higher prices of goods and services due to increased cost of production.
5. **Exchange Rates:** Exposure to foreign markets are based on the dollar value. Fluctuations in the exchange rate have an impact on the rate of inflation.

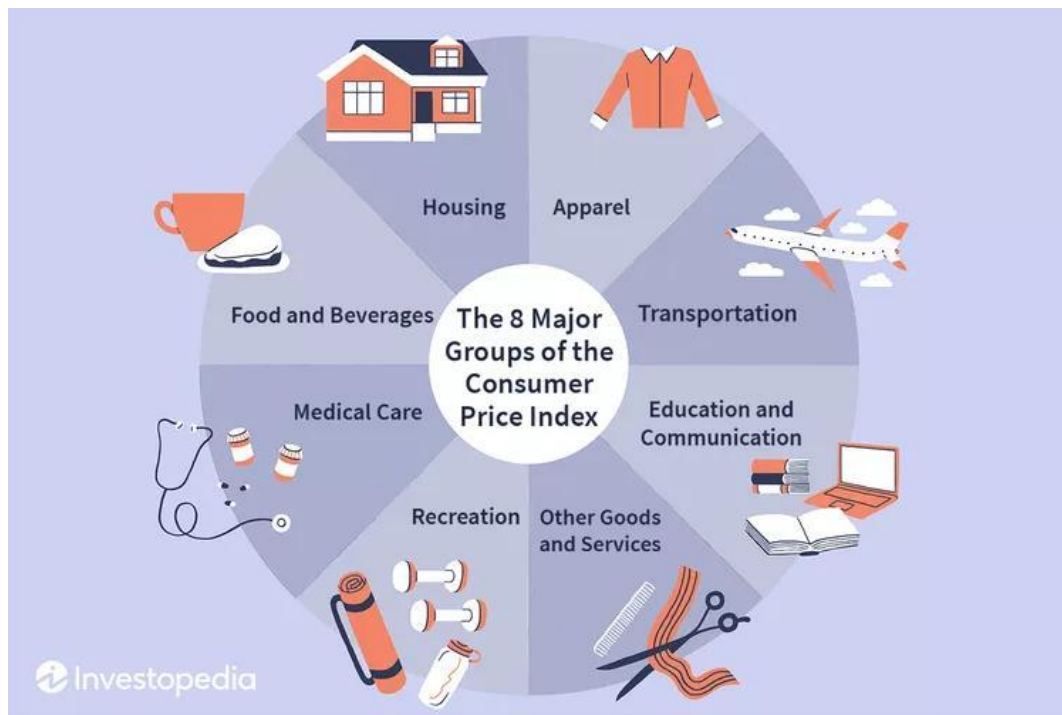
What is the Inflation Rate Formula?

Inflation rate formula is the difference between initial CPI and final CPI divided by initial CPI. The result then multiplied by 100 gives the inflation rate.

Rate of Inflation = $(\text{Initial CPI} - \text{Final CPI}) / \text{Initial CPI}$

*100 CPI= Consumer Price Index

A **consumer price index (CPI)** is a price index, the price of a weighted average market basket of consumer goods and services purchased by households. Changes in measured CPI track changes in prices over time.



CPI Categories by Weight as of February 2023

Group	Weight
Housing	34.4%
Food	13.5%
CPI Categories by Weight as of February 2023	
Transportation	5.7%
Commodities	21.3%
Health Care	6.6%
Energy	7.1%
Education	4.9%
Other Expenses	6.5%
Total Expenses	100%

Source: Bureau of Labor Statistics

What Is Deflation?

Deflation is a phenomenon, which is exactly the opposite of inflation. When deflation occurs, the prices of goods and services fall which in turn increases the purchasing power of the money. It also means that more goods and services can be bought with the same amount of money.

Let us look at some of the points of difference between inflation and deflation.

Inflation	Deflation
Definition	
Inflation is defined as the increase in the price levels of goods and services in an economy	Deflation is termed as the decrease in price levels of goods and services in an economy
Impact on demand	
Demand for products and services increase in inflation	Demand for products and services decrease in deflation.
Impact on National Income	
No impact on national income	National income declines as a result of deflation
Consequences seen	
Distribution of income is not equal as a result of inflation	There is a rise in level of unemployment in the nation as a result of deflation
Is it beneficial?	
Moderate levels of inflation is considered good for the economy	Calculated based on only the amount that is availed
Impact on Purchasing Power of Money	
Decreases the purchasing power of money	Increases the purchasing power of money

Ans.6 Explain about the Joint Stock Company: -

The simplest way to describe a joint stock company is that it is a business organisation that is owned jointly by all its shareholders. All the shareholders own a certain amount of stock in the company, which is represented by their shares.

Example of Joint Stock Company

Few examples are mentioned below.

- Indian Oil Corporation Ltd.
- Tata Motors Ltd.
- Reliance Industries Ltd.

Features of Joint Stock Company: -

1. **Separate Legal Entity** – A joint stock company is an individual legal entity, apart from the persons involved. It can own assets and can because it is an entity it can sue or can be sued. Whereas a partnership or a sole proprietor, it has no such legal existence apart from the person involved in it.
2. **Number of Members** – For a public limited company, there can be an unlimited number of members but minimum being seven. For a private limited company, only two members. In general, a partnership firm cannot have more than 10 members in one business.
3. **Limited Liability** – In this type of company, the liability of the company's shareholders is limited. However, no member can liquidate the personal assets to pay the debts of a firm.
4. **Transferable share** – A company's shareholder without consulting can transfer his shares to others. Whereas, in a partnership firm without any approval of other partners, a partner cannot move his share.
5. **Incorporation** – For a firm to be accepted as an individual legal entity, it has to be incorporated. So, it is compulsory to register a firm under a joint stock company.

Merits of Joint Stock Company:-

The important advantages of a joint stock company are as follows:

1. **Limited Liability:** The liability of the members of a company is limited to the extent of the share contributed by them in the company. If a company faces a loss, the shareholders of the company do not have to sell off their personal property for repayment. This advantage of a joint stock company attracts people to invest money in the Company Form of Business.
2. **Transfer of Interest:** As the shares of a company are transferrable and can be easily bought and sold in the market, it brings liquidity of investment in the company. The shareholders can anytime convert their share investment into cash and can use that amount to buy the shares of another company.
4. **Growth and Expansion:** A company has more scope for expansion and growth because it has large financial resources and high profit rates. It means that if a company has retained profits, it can easily use that amount for its growth and expansion.

5. Efficient Management: Every business requires specialised people and experts for better performance and results. As a company has huge funds at its disposal, it can easily hire experts to perform various business activities, and can efficiently improve its working and performance.

6. Large Amount of Capital: As a company can issue shares to the general public, it grabs the biggest advantage of large capital.

Demerits of Joint Stock Company: -

The major limitations of a joint stock company are as follows:

1. Complexity in Formation: The process of the formation of a company is quite complicated and lengthy. It requires the completion of various formalities, and various different documents have to be prepared and submitted. It also requires obtaining different kinds of permissions. For all these activities, a company has to hire experts who charge high fees for the same. Besides, registration fees have to be paid to the registrar of companies.

2. Lack of Secrecy: According to the Companies Act, 1956 every company has to share various information about it with the registrar of companies, which is made available to the general public also. This compulsion of sharing information makes it difficult for the company to maintain secrecy about its operations.

3. Impersonal Work Environment: As a company is managed by hired professionals and experts, instead of its owners, and the professionals get a salary in return, there is no direct relationship between the efforts and reward of the business activities.

4. Delay in Decisions: The important decisions in a company are taken after consulting with different people or discussing in the board meeting, which is a lengthy process.

5. Conflicts in Interests: Various groups of people such as debenture holders, shareholders, directors, employees, managers, etc., are involved in the operations of a company, and every group has its own interest. It is not possible that every group's interest aligns with a decision, which can result in conflicts in interests between different groups.

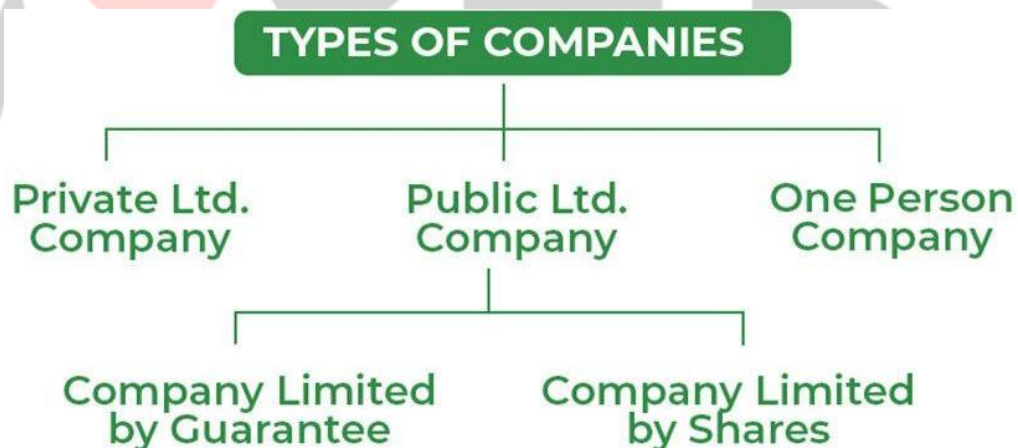
Types of Joint Stock Company

The joint stock company is divided into three different types.

- **Chartered Company** – A firm incorporated by the king or the head of the state is known as a chartered company.
- **Statutory Company** – A company which is formed by a particular act of parliament is known as a statutory company. Here, all the power, object, right, and responsibility are all defined by the act.
- **Registered Company** – An organisation that is formed by registering under the law of the company comes under a registered company.

Types of Companies:-

Based on ownership, there are three categories of companies; viz., **Private Ltd. Company**, **Public Ltd. Company**, and **One Person Company**.



1. Private Ltd. Company

According to Section 2(68) of the Indian Companies Act, 2013, a Private Company by its article is a company which:

4. Restricts the rights to transfer its shares.

5. Has a minimum 2 and a maximum of 200 members, excluding the present and past employees.
6. Prohibits any invitation to the public to subscribe for a company's securities. Examples of Private Limited Companies

The companies that run Flipkart, Ola, Snapdeal, Carat Lane, and Zoom Car are all private entities.

2. Public Ltd. Company

A company which is not a private company is a Public Ltd. Company; however, a private company which is a subsidiary of a public company is treated as a public company. According to the Companies Act, a public company is a company which :

1. Has minimum 7 members and no limit on the maximum number of members.
2. Has no restriction on the transfer of securities.
3. Is not prohibited to invite the general public to subscribe to its securities.

TOP 10 PUBLIC LIMITED COMPANIES IN INDIA

Check out the list of Top 10 Public Limited Companies in India-

1. **Bharat Heavy Electricals Ltd. (BHEL)**
2. **Bharat Petroleum Corporation Ltd (BPCL)**
3. **Coal India Ltd.**
4. **Hindustan Corporation Petroleum Ltd.**
5. **Indian Oil Corporation Ltd.**
6. **NTPC Ltd.**
7. **Oil and Natural gas Corporation Ltd. (ONGC)**
8. **Punjab National Bank**
9. **State Bank of India**
10. **Steel Authority of India Ltd.**

A. Company Limited by Guarantee

A company with no shareholders, which is owned by the members, also known as **guarantors** is known as a Company Limited by Guarantee or a Guarantee Company. The guarantors agree to pay a nominal amount to the company if it gets wound up. This form of business is usually used for non-profit organisations.

B. Company Limited by Shares

A company with members whose liability is limited to the amount unpaid (if any) on the shares held by them is known as a Company Limited by Shares.

3. One Person Company

According to the Companies Act, 2013, apart from a public or private limited company, one can also form a new type of entity, **One Person Company (OPC)**. As per section 3(1) of the Companies Act 2013, a One Person Company is a company which has only one person as its member. Rule 3(1) states that only a natural person who is also an Indian citizen and resident is eligible to incorporate One Person Company.

Famous One Person Company in India

One person company has been preferred over another form of investment incorporations by a lot of budding freelancers and entrepreneurs primarily because of minimal legal compliances and maximum taxation benefits a few examples being

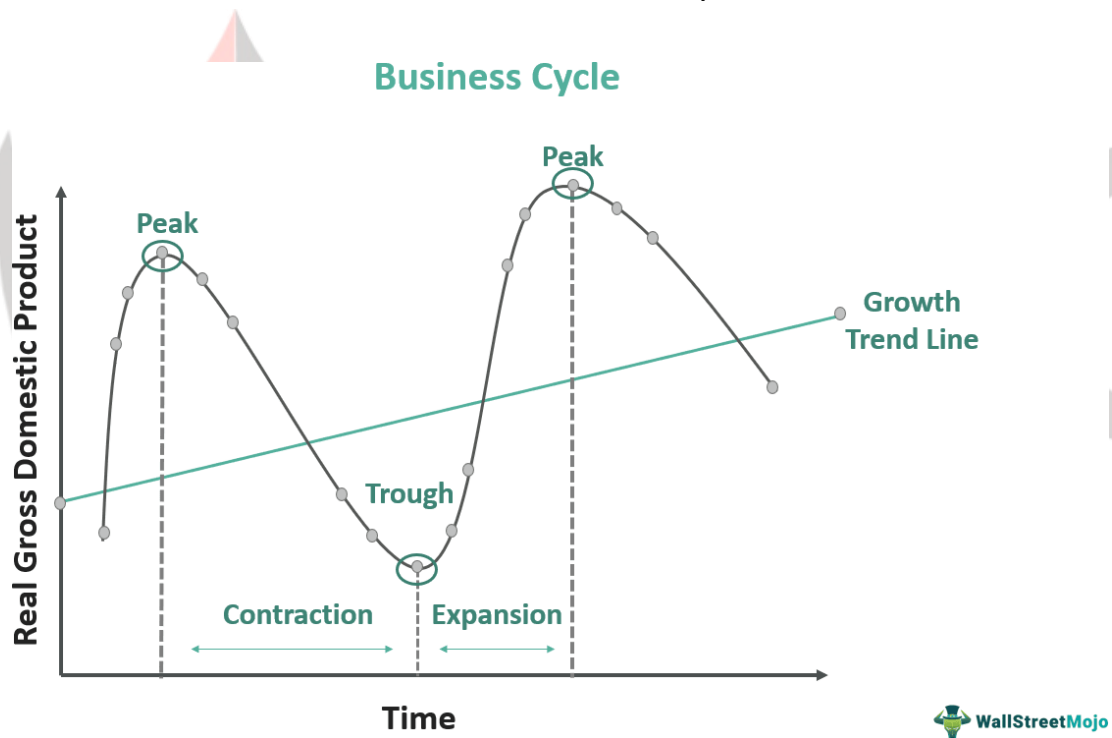
- BHAIKAV BUILDERS PRIVATE LIMITED (OPC) is a Private (One Person Company) in Delhi and been in existence for more than 20 years. it deals in buying, selling, renting and operating of self-owned or leased real estate
- DIGITEX FABRICS OPC PRIVATE LIMITED is a Private (One Person Company) incorporated on 05 May 2015. It is involved in Spinning, weaving and finishing of textiles.

Ans.7 What is the Business Cycle? Discuss about the various Phases and features of Business Cycle.

The Business Cycle refers to the vast Economic fluctuations in trade, production, and general Economic activities. The Business Cycle is also known as the boom-bust Cycle or Economic Cycle. If we look at it conceptually then, the Business Cycle refers to the up and down movements of the GDP and refers to widespread expansions and contractions in the level of Economic booming and activities.

What are the Phases of a Business Cycle?

As mentioned earlier, there are four different phases of the Business Cycle; all these phases have different features of the Business Cycle.



1. **Expansion:** Expansion is characterized by employment increase, Economic growth, and rise in prices.
2. **Peak:** Peak is considered to be the highest point in the Business Cycle. A Business Cycle is said to have reached a peak when there is maximum output, employment is full or near to full employment, and inflationary prices are somewhat evident.

3. **Contraction:** Once the peak is reached, the Economy usually enters into the contraction phase. In this phase, the growth slows down and unemployment increases.
4. **Trough:** The contraction then hits the trough point, and it is at this time that the Economy has its bottom. Once again, the Economy will hit bottom, and the next phase of expansion and contraction will start.

What are the Features of a Business Cycle?

There are several features of a Business Cycle. Let us take a look at five features of a Business Cycle.

1. **Occurs Periodically:** The different phases of a Business Cycle occur from time to time. Although, at certain times, these periods will vary according to the Economic conditions of the industry.
2. **Synchronous:** Another advantageous and prominent feature of the Business Cycle is that it is synchronic. The features of a Business Cycle are not restricted to a single firm or industry. They originate in a free Economy and are prevalent.
3. **Major Sectors are affected:** It's been noticed that fluctuations occur not only at the level of production but also in other variables such as employment, consumption, investment, rate of interest, and price level.
4. **Profit Variation:** Another significant feature of the Business Cycle is that the profits fluctuate more than any other income source.
5. **Worldwide Impact:** Business Cycles are international in nature.

Ans.8 Elaborate the Features and Advantages of Double Entry? What Is an Example of Double Entry?

An example of double-entry accounting would be if a business took out a \$10,000 loan and the loan was recorded in both the debit account and the credit account. The cash (asset) account would be debited by \$10,000 and the debt (liability) account is credited by \$10,000. Under the double-entry system, both the debit and credit accounts will equal each other.

Features of Double Entry

- **Two Parties:** Two parties are involved, one is the receiver, and another is the giver. The receiving party is debited, and another party is credited. For example, A purchases goods from B, where A is a receiver party, and B is a giver party.
- **Equal Effect:** Each transaction should have an equal financial effect. The debit amount should be equal to the credit amount.
- **Separate Legal Entity:** This accounting system records the transaction separate from its owners.
- **Debit and Credit:** There are two aspects for recording any transaction, the Debit aspect, and the Credit aspect.

Principle of Double Entry

Double-entry is based on a simple principle, that for every **debit**, must have equal and opposite credit. There should be at least two accounts involved in any transaction.

Debit Side = Credit Side

The double-entry is based on the debit and credit accounts of the transaction. So, we need to understand what account kind of debits and what credits.

Example of Double Entry

Here are a few transactions for which Journal Entries are to be recorded. Record the entries in the Books of A Limited.

A Limited Purchases Goods worth Rs. 2,500 from B Limited on Credit.

Particulars	Debit	Credit
Purchase A/c	Rs. 2,500	
B Limited A/c		Rs. 2,500

A Limited makes a payment for the Goods next Month.

Particulars	Debit	Credit
B Limited A/c	Rs.2,500	
Cash A/c		Rs.2,500

A Limited Purchases Machinery worth Rs.30,000 by paying cash:

Particulars	Debit	Credit
Machinery A/c	Rs.30,000	
Cash A/c		Rs.30,000

A Limited received Rent on Building Rs.1,500:

Particulars	Debit	Credit
Cash A/c	Rs.1,500	
Rent Received A/c		Rs.1,500

Difference between Double Entry and Single Entry

Basis	Double Entry System	Single Entry System
Meaning	It is the method of accounting where the dual aspect of the transaction is recorded	It is the method of accounting where only one side of transaction is recorded
Nature	It is a complex form of accounting.	It is a simple form of accounting.
Accuracy	It provides more accurate financial results	Since it record only one side of transaction hence less accuracy.



Basis	Double Entry System	Single Entry System
Scale of Business	Preferable for large scale business	Preferable for small scale business
Level of Completion	It is a complete form of accounting	It provides incomplete results
Detection of Errors	Errors can be detected easily	Difficult to detect errors as only one side of transaction is recorded
Cost	Higher cost as skilled staff is required to maintain the book of accounts	Less cost as it is a simple form of accounting

Advantages:-

- **Modern and Scientific:** Double entry is a scientific and systematic system of recording and maintaining books of accounts. There are the Rules and Principles which have to be followed rigorously.
- **Complete System of Accounting:** This form of accounting records both aspects of a transaction; hence, it is a complete form of accounting.
- **Fewer Errors:** There are fewer chances of errors as both the debit and credit sides of the transaction have to be equal.
- **Fraud Prevention:** This accounting system helps in the prevention and early detection of fraud.
- **Decision Making:** A complete set of books helps in decision making for management, investors, creditors, **and auditors.**

Disadvantages:-

- **Complex:** This is a more complex form of accounting. The person must know the **rules of accounting** to ensure the books are maintained correctly.
- **High Cost:** Since this **system of accounting** required skilled personnel hence the cost of hiring them would also be high.

- **Not suitable for Small Business:** Small businesses with fewer transactions would not find this **method of accounting** suitable for them.

Ans.9 What Is Ratio Analysis? Explain the Various types of Ratios with some Examples.

Ratio analysis is a quantitative method of gaining insight into a company's liquidity, operational efficiency, and profitability by studying its financial statements such as the balance sheet and income statement.

Importance, Significance, and Merits of Ratio Analysis

The main points of importance are as follows:

1. Test of solvency. Ratios can illuminate the solvency of a firm. For example, when the **ratio of current assets to current liabilities** is increasing, this indicates sufficient **working capital**. Thus, creditors can be paid easily.

2. Helpful in decision-making. The main aim of financial statements is to inform users about the financial position of the company, as well as to serve as a decision-making aid for managerial personnel.

3. Helpful in financial forecasting and planning. Ratios are critical in **financial planning** and forecasting. For example, if a firm's **current ratio** is 5:1, this means that **capital** is blocked up. As the ideal ratio is 2:1, we have 5:1, meaning that \$3 is unnecessarily blocked.

4. Useful in discovering profitability. Ratios are also useful when comparing the **profitability** of different companies. Present and past ratios can be compared, for example, to discover trends in the historical and future performance of companies.

5. Liquidity position. With the use of ratio analysis, meaningful conclusions can be obtained about the sound **liquidity** position of the firm. A firm's liquidity position is sound if it can pay its debts when these are due for payments.

6. Useful for operating efficiency. From a management perspective, ratios

enable managers to measure the efficiency of **assets**. When sales and their contribution to **net profit** increase every year, this is a test of higher efficiency.

7. Business trends. Ratio analysis can expose trends that managers may use to take corrective actions.

8. Helpful in cost control. Ratios are useful to measure performance and facilitate cost control.

9. Helpful in analyzing corporate financial health. Ratio analysis can provide information about **liquidity**, solvency, profitability, and capital gearing. Thus, they are valuable for learning about financial health.

Limitations of Ratio Analysis

Although ratios are useful tools, they should be used with the utmost care. This is because they can suffer from drawbacks and limitations, including:

1. Need for technical knowledge. Ratios are quantitative and not **qualitative** indicators. Thus, to use them, one needs some knowledge of quantitative analysis.

2. Lack of reliable data. When figures are incorrect (e.g., value of closing stock is overstated), ratios will give misleading results.

3. Different basis. Different methods are available for the **valuation** of closing stock: **LIFO** and **FIFO**. In both, profit will differ. Similarly, profit has different meanings.

For example, some companies may take profit before **tax** and **interest**, while others may take profit after tax and interest. Similarly, different methods of depreciation will show different amounts of profit.

4. Different accounting policies. Different firms follow different policies with regard to **depreciation** (e.g., **fixed installments** or **diminishing balance method**, or stock valuation). Therefore, unless adjustments for profit are

made, profit will not be comparable.

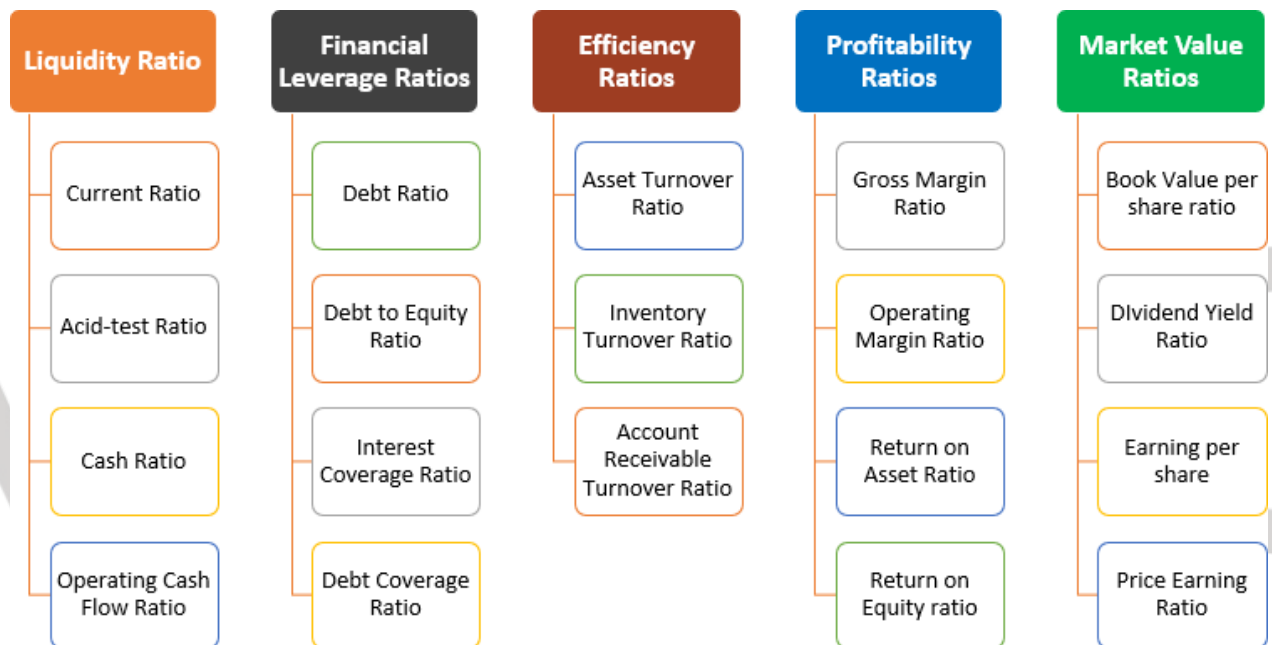
5. Effect of price level change. When ratios are calculated, no thought is given to inflationary measures that are responsible for changes in price. Thus, the utility of ratio analysis becomes questionable in these cases.



6. Bias. Ratios are only tools. They depend on the user for practical shape. For example, profit has different meanings, including **EBIT (earnings before interest and taxes)**. Thus, personal opinion differs from business to business.

7. Lack of comparison. Different firms adopt different procedures, records, objectives, and policies. Due to this, comparisons become complex.

8. Evaluation. There are different tools for ratio analysis. The question of which tool to use in a particular situation depends upon the skill, training, knowledge, and expertise of the analyst.



TYPES OF RATIOS: -

The ratios can be classified into following four broad categories:

1. Financial position Ratios
 - A. Liquidity Ratios /short term financial position Ratios
 - B. Capital Structure/Long term financial position Ratios
2. Activity Ratios/ Turnover Ratios
3. Profitability Ratios
4. Investors/Other Ratios

1. Financial position Ratios

A. Liquidity Ratios /short term financial position Ratios

- a) **Current Ratios:** The Current Ratio is one of the best known measures of financial strength.

Current Ratio = Current Assets / Current Liabilities

Where,

Current Assets = Inventories + Sundry Debtors + Cash and Bank Balances + Receivables/ Accruals + Loans and Advances + Disposable Investments

Current Liabilities = Creditors for goods and services + Short-term Loans + Bank Overdraft + Cash Credit + Outstanding Expenses + Provision for Taxation + Proposed Dividend + Unclaimed Dividend

A generally acceptable current ratio is 2 to 1. But whether or not a specific ratio is satisfactory depends on the nature of the business and the characteristics of its current assets and liabilities.

- b) **Quick Ratios:** The Quick Ratio is sometimes called the "acid-test" ratio and is one of the best measures of liquidity.

Quick Ratio or Acid Test Ratio = Quick Assets/ Current Liabilities Where,

Quick Assets = Current Assets - Inventories & prepaid Expenses

Current Liabilities = As mentioned under Current Ratio.

The Quick Ratio is a much more conservative measure than the Current Ratio. It helps answer the question: "If all sales revenues should disappear, could my business meet its current obligations with the readily convertible 'quick' funds on hand?"

Quick Assets consist of only cash and near cash assets. Inventories are deducted from current assets on the belief that these are not 'near cash assets'. But in a seller's market inventories are also near cash assets.

An acid-test of 1:1 is considered satisfactory unless the majority of "quick

assets" are in accounts receivable, and the pattern of accounts receivable collection lags behind the schedule for paying current liabilities

Question

Calculate Current Ratio, Quick Ratio and Debt to Equity Ratio from the figures given below:

Particulars	₹
Inventory	30,000
Prepaid Expenses	2,000
Other Current Assets	50,000
Current Liabilities	40,000
12% Debentures	30,000
Accumulated Profits	10,000
Equity Share Capital	1,00,000
Non-current Investments	15,000

Solution

(i)

$$\begin{aligned}\text{Current Assets} &= \text{Inventory} + \text{Prepaid Expenses} + \text{Other Current Assets} \\ &= 30,000 + 2,000 + 50,000 = 82,000\end{aligned}$$

$$\text{Current Liabilities} = 40,000$$

$$\begin{aligned}\text{Current Ratio} &= \frac{\text{Current Assets}}{\text{Current Liabilities}} \\ &= \frac{82,000}{40,000} = 2.05 : 1\end{aligned}$$

(ii)

$$\begin{aligned}\text{Liquid Assets} &= \text{Current Assets} - \text{Inventory} - \text{Prepaid Expenses} \\ &= 82,000 - 30,000 - 2,000 = 50,000\end{aligned}$$

$$\text{Quick Ratio} = \frac{\text{Liquid Assets}}{\text{Current Liabilities}} = \frac{50,000}{40,000} = 1.25 : 1$$

(iii)

$$\text{Long-term Debts} = 12\% \text{ Debentures} = 30,000$$

$$\begin{aligned}\text{Equity} &= \text{Accumulated Profits} + \text{Equity Share Capital} \\ &= 10,000 + 1,00,000 = 1,10,000\end{aligned}$$

$$\begin{aligned}\text{Debt-Equity Ratio} &= \frac{\text{Long-Term Debts}}{\text{Equity}} \\ &= \frac{30,000}{1,10,000} = 0.27 : 1\end{aligned}$$

What is working capital?

"Working capital" is the money you need to support short-term operations. It is this focus on the short term that distinguishes working capital from longer-term investments in fixed assets or R&D.

Working Capital Ratio = Current Assets ÷ Current Liabilities

Let us calculate working Capital from the images above.

Here, Current Assets = Cash and Cash Equivalents + Accounts Receivables + Inventories + Other Current Assets

- Current Assets (2015) = \$970 + \$1,427 + \$1,180 + \$807 = \$4,384

Current Liabilities = Notes and loans payable + Current portion of long term

debt + Accounts Payable + Accrued Income Taxes + Other Accruals

- Current Liabilities (2015) = \$4 + \$298 + \$1,110 + \$277 + \$1,845 =

\$3,534 Working Capital (2015) = Current Assets (2015) – Current

Liabilities (2015)

- Working Capital (2015) = \$4,384 – \$3,534 = \$850
- Working Capital Ratio (2015) = \$4,384 / \$3,534 = 1.24
- This ratio is also known as Current Ratio.

Earning Per share Ratio:-

What is Earnings Per Share (EPS)?

Earnings Per Share (EPS) is a financial metric calculated by dividing the Net income by the total number of outstanding common shares. Investors use EPS to assess a company's performance and profitability before investing. Higher EPS means the company is more profitable.

Earnings Per Share Formula

$$\text{EPS} = \frac{\text{Net Income of the Company}}{\text{Average Outstanding Shares of the Company}}$$

$$\text{EPS} = \frac{\text{Net Income of the Company} - \text{Dividend to Preferred Shareholders}}{\text{Average Outstanding Shares of the Company}}$$

Example #1

Hit Technology Inc. has the following information –

- The net income for the year-end 2017 – \$450,000
- The preferred dividends paid in 2017 – \$30,000
- At the beginning of the year 2017, the common shares outstanding were 50,000 shares. In the middle of the year, Hit Technology Inc. issued another 40,000 common shares.

Find out the earnings per share of Hit

Technology Inc. Solution:-

Here's the calculation –

- Weighted average number of common shares = $(50,000 * 1) + (40,000 * 0.5) = 50,000 + 20,000 = 70,000$ shares.

Now, we will find out the EPS formula –

- EPS formula = $(\text{Net Income} - \text{Preferred Dividends}) / \text{Weighted Average Number of Common Shares}$
- Or. EPS formula = $(\$450,000 - \$30,000) / 70,000$
- Or, EPS = $\$420,000 / 70,000 = \6 per share.

Debtors Turnover Ratio:

This ratio measures the efficiency with which Accounts Receivable are being managed, hence it is also known as 'Accounts Receivable Turnover ratio'. The ratio shows the equation between credit sales (cash sales are not taken into consideration) and the average debtors of a firm. The formula is as below:-

Example of Debtors Turnover Ratio

Let's consider that a company, ABC, has made net credit sales of Rs. 200,000 for a year with a sales return amount of Rs. 20,000. The accounts receivables amount is Rs. 25,000.

To find out net credit sales, follow this debtors turnover ratio

formula: Net Credit Sales = Sales - Sales returns-Cash sales

So, Rs. 200,000 - INR 20,000 = INR 180,000

You already know that the average accounts receivable is Rs.

25000. Considering the debtors turnover ratio formula, the turnover ratio is: $180,000/25,000 = 7.2$

Hence, the debtors turnover ratio amount is 7.2.

Solved Examples for You

Q: Calculate Debtors Turnover Ratio and Average Collection Period (in days) from the following.

Total Sales – 6,00,000

Cash Sales – 20% of Total sales

Trades Receivable at beginning of the year- 80,000

Trades Receivable at the end of the year- 1,60,000

Solution: From the given information

Credit Sales = 80% of Total Sales = 80% of 6,00,000 = 4,80,000

Average Debtors = (OpeningDebtors+ClosingDebtors)/2

= (80,000+1,60,000)/2 = 1,20,000

Debtors' Turnover ratio = Credit Sales/Average Debtors

= 480000/120000

= 4 times

Average Collection Period = $\frac{\text{Number of days/weeks/months}}{\text{Debtors turnover Ratio}}$



$365/4 = 91.25$ days 92 days

$$\text{Creditors' turnover ratio} = \frac{\text{Credit Purchases}}{\text{Average Creditors}}$$

$$\text{Average creditors} = \frac{\text{opening creditors} + \text{closing creditors}}{2}$$

High and Low Creditor's Turnover Ratio

A high ratio may indicate

- Low credit period available to the business or early payments made by the business.
- The company may operate majorly on the cash basis.
- The company is not availing full credit period.

A low ratio may indicate

- Creditors are not paid in time.
- Increased credit period is allowed to the business.

Example – Payables Turnover Ratio

Ques. Calculate creditor's turnover ratio from the information provided

below; Total Purchases – 5,00,000

Cash Purchases – 2,00,000

Creditors (Beginning of period) – 50,000 & Creditors (End of period) – 1,00,000

Ans.

Creditor's turnover ratio or Accounts payable turnover ratio = **(Net Credit Sales/Average Trade Receivables)**

$$\begin{aligned} \text{Net Credit Purchases} &= \text{Total Purchases} - \text{Cash Purchases} \\ &= 5,00,000 - 2,00,000 \end{aligned}$$

$$\text{Net Credit Purchases} = 3,00,000$$

Average Trade Payables = (Opening Trade Payables + Closing Trade Payables)/2

= (50,000 + 1,00,000)/2

= 75,000

Ratio = (3,00,000/75,000) = 4/1 or 4:1

What is gross profit?

Gross profit is the profit a business makes after subtracting all the costs that are related to manufacturing and selling its products or services. You can calculate gross profit by deducting the cost of goods sold (COGS) from your total sales.

What is net profit?

Net profit is the amount of money your business earns after deducting all operating, interest, and tax expenses over a given period of time. To arrive at this value, you need to know a company's gross profit. If the value of net profit is negative, then it is called net loss.

How to calculate gross and net profit?

You can calculate both gross and net profit using your income statement. An income statement shows your company's total revenue and cost of goods sold, followed by the operating expenses, interest and taxes.

Question

From the following calculate

(a) Net Profit Ratio

(b) Operating Profit Ratio

S. No.	Items Amount	Rs.
--------	--------------	-----

1.	Revenue from Operations (Sales):-	2,00,000
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2.	Gross Profit-	75,000
----	---------------	--------

3.	Office Expenses-	15,000
----	------------------	--------

4.	Selling Expenses-	26,000
----	-------------------	--------

5.	Interest on Debentures-	5,000
----	-------------------------	-------

6.	Accidental losses-	12,000
----	--------------------	--------

7.	Income from Rent-	2,500
----	-------------------	-------

8.	Commission received-	2,000
----	----------------------	-------

Solution

(a) Net Profit Ratio = $\frac{\text{Net Profit}}{\text{Revenue from Operations}} \times 100$
Net Profit = Gross Profit - Indirect Expenses & Losses + Other Incomes

Indirect Expenses and Losses = Office Expenses + Selling Expenses + Interest on Debentures + Accidental Losses

= Rs. 15,000 + Rs. 26,000 + Rs. 5,000 + Rs. 12,000

= Rs. 58,000

Other Incomes = Income from Rent + Commission Received

= Rs. 2,500 + Rs. 2,000 = Rs. 4,500

Net Profit = Rs. 75,000 - Rs. 58,000 + Rs. 4,500 = Rs. 21,500

Revenue from Operations = Rs. 2,00,000

$\therefore$ Net Profit Ratio = $\frac{21,500}{2,00,000} \times 100 = 10.75$

(b) Operating Profit Ratio = $\frac{\text{Operating Profit}}{\text{Revenue from Operations}} \times 100$

Operating Profit = Gross Profit + Operating Income - Operating Expenses

Operating Income = Commission Received

= Rs. 2,000

Operating Expenses = Office Expenses + Selling Expenses

= Rs. 15,000 + Rs. 26,000 = Rs. 41,000

∴ Operating Profit = Rs. 75,000 + Rs. 2,000 - Rs. 41,000 = Rs. 36,000

Revenue from Operations = Rs. 2,00,000

∴ Operating Profit Ratio = $\frac{\text{Rs. 36,000}}{\text{Rs. 2,00,000}} \times 100 = 18\%$

Practical Problems with Answers:-

Balance Sheet Ratios:-

1. Current Ratio = Current Assets/Current Liabilities

2. Liquid Ratio = Quick (Liquid) Assets/Quick (Liquid) Liabilities

- Liquid assets are those which are readily converted into cash and will include cash/ bank balances, bills receivable, sundry debtors and short term investments. Inventories and Prepaid Expenses are not included in liquid assets.
- Liquid Liabilities includes all items of current liabilities except Bank Overdraft.

3. Proprietary Ratio = Proprietors funds/Total Assets

- Proprietor fund = Share capital(Equity & Pref.) + Retained earnings (less loss if any) -Fictitious assets
- Total Assets = Fixed Assets + Current Assets-Fictitious assets

Q.

	Rs.	Assets	Rs.
Liabilities			
Equity Share Capital	5,00,000	Land & Building	1,00,000
Preference share	2,00,000	Machinery	4,00,000

capital			
General Reserve	1,00,000	Furniture	50,000
Secured Loan	3,00,000	Inventory	3,00,000
Sundry Creditors	1,00,000	Sundry Debtors	3,00,000
		Cash/Bank Balance	50,000
	12,00,000		12,00,000

Calculate Following Ratios from the above balance sheet:

1. Current Ratio
2. Liquid Ratio
3. Proprietary Ratio

Solution:

1. Current ratio	$= \frac{\text{Current assets}}{\text{current liabilities}}$ $\text{Current assets} = \text{inventory} + \text{s.debtors} + \text{cash balance}$ $(3,00,000) + \text{s.debtors}(3,00,000) + \text{cash balance}(50,000) = 6,50,000$ $\text{Current liabilities} = \text{S. Creditors} = 1,00,000$ $= \frac{6,50,000}{1,00,000}$ $= 6.5:1$
2. Liquid ratio	$= \frac{\text{liquid assets}}{\text{liquid liabilities}}$ $\text{liquid assets} = \text{s.debtors}(3,00,000) + \text{cash balance}(50,000)$ $= 3,50,000$ $\text{liquid liabilities} = \text{S.Creditors} = 1,00,000$ $= \frac{3,50,000}{1,00,000}$ $= 3.5:1$
3. Proprietary Ratio	$\text{Proprietors fund} / \text{total assets}$
	$\text{Proprietor fund} = \text{Share capital (Equity \& Pref.)} + \text{Retained}$

	earnings (less loss if any) -Fictitious assets
	= 5,00,000 + 2,00,000 + 100,000 = 8,00,000
	Total Assets = Fixed Assets + Current Assets- Fictitious
	Assets
	= 12,00,000
	= 800,000/12,00,000
	= 0.66 : 1

Ans.10 Discuss the Determinants and Method of demand forecasting?

Determinants and Method of demand forecasting: -

some common demand forecasting methods, including:

1. Statistical Method

This approach uses historical data to identify patterns and trends that can be used to make predictions about future demand. This method is best suited for businesses with a long sales data history. It will be less effective for new businesses or have experienced significant changes in demand over time.

The statistical method can be divided into two sub-categories:

- **Trend Projection:** In this approach, demand is forecasted by extrapolating past demand data into the future. This method can be used when demand grows at a constant rate or when there is a clear trend in the data.
- **Regression Analysis:** This approach uses historical data to identify relationships between different factors and demand. The sourced information can be utilised to make predictions about demand in the future. This method is best suited for businesses with a large amount of data that can be analysed.

2. Market Research/Surveying Method

Before thinking about demand forecasting, you need to understand your target market. This means conducting extensive market research to get an idea of your

target customers, their needs and wants, and what factors influence their buying decisions. One of the best ways to do this is through surveying. A survey can be either done by the company itself or by hiring a market research firm. Either way, surveys are a great way to get reliable data that you can use to make informed decisions about demand forecasting.

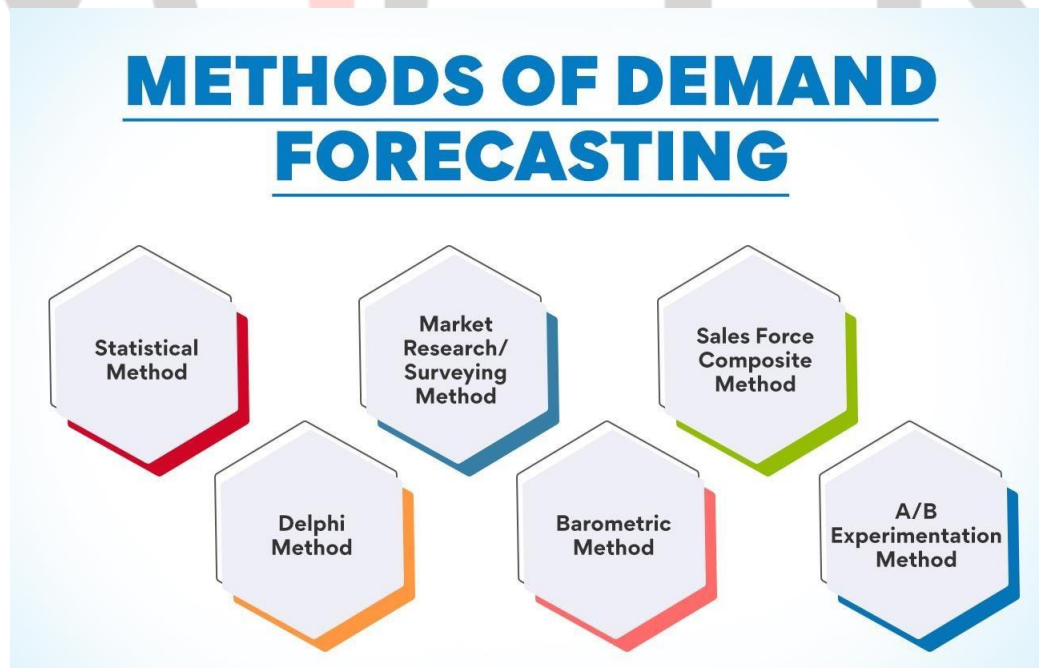
3. Sales Force Composite Method

The sales force composite method is a technique that relies on the collective knowledge of your sales team. Essentially, you ask your salespeople to estimate how much demand they think there will be for your product or service in the future. All the information is compiled and used to generate a demand forecast.

4. Delphi Method

It is a demand forecasting technique that relies on the opinions of experts. The Delphi method is used to generate forecasts by asking a panel of experts their opinions on the future demand for a product. The experts' responses are then collated, and an average demand forecast is generated.

The Delphi method is ideal when there is little historical data available on which to base a demand forecast. The Delphi method can be used to generate forecasts for both short-term and long-term demand.



5. Barometric Method

This demand forecasting method uses historical sales data to predict future demand. It's based on the theory that demand for a product is influenced by previous demand for that product. To forecast demand using this method, you'll need sales data for a similar product from past years. You can then use this data to identify patterns and trends to help predict future demand. The barometric method is best suited for products with stable demand. If demand for your product fluctuates significantly from year to year, this method may not be as accurate.

6. A/B Experimentation Method

This method is all about experimentation. You create two demand forecasts, one using your current methods and assumptions and a second demand forecast with different techniques and assumptions. Then, you compare the results of both demand forecasts to see which one is more accurate. It is an effective method when you have a limited amount of historical data to work with.

Ans.11 Describe the Factors that Affect Supply and Exceptions to Law of Supply: -

1. Price and demand forecasts: -

Many businesses base their production plans on forecasts of future demand and pricing, not just on what customers are currently buying.

2. Production costs: -

The law of supply assumes that companies can increase profits by selling more goods or services when prices rise, which provides them with an incentive to increase the supply. But if the price rises reflect increased production costs, that may not be true.

3. Competition:-

- New suppliers may enter the market even if prices are not increasing and

demand is stable.

4. Technology:-Technology can enable companies to make and sell more products at a lower cost, thus increasing the available supply.

5. Transportation:-Transportation delays or rising shipping costs can affect a company's ability to increase its supply of goods. If goods can't move from warehouses to retail shelves, they can't be purchased by customers and don't count toward the market supply.

6. Availability of raw materials and labor:- A business may want to increase the supply of a product but unable to do so because it can't purchase the raw materials or hire the people required to produce it.

7. Government regulations and subsidies influence supply in some industries:- Companies must meet strict regulatory requirements when introducing certain healthcare products, for example, which can limit the supply of these products regardless of the demand. On the other hand, government subsidies support the supply of some local transportation services.

8. Weather and natural disasters:-For many agricultural goods, the weather has a major impact on supply. A dry season or flooding can greatly reduce crop yields.

9. Comparable goods:- A change in the supply of one good can affect the supply of other goods. For example, if the market price of corn increases, farmers may dedicate more land to growing corn. As a result, they use less land for growing squash, so the supply of squash decreases.

Exceptions to Law of Supply:-

1. Shift in business plan:- If a business is shifting its market focus and plans to

cease production of some products, it may temporarily increase the supply of those products at a low price to eliminate any remaining stock and raw materials. A business may also use this approach as an emergency measure if it needs cash in a hurry.

2. Monopoly: -When there's only a single supplier of a good or service, the company may be able to increase or decrease its supply or pricing irrespective of external factors.

3. Competitive pricing:-In a highly competitive market, businesses may increase the supply of their products while reducing the price to capture market share.

4. Expiring or dated goods:- If perishable goods near their expiration date, a business may increase their supply early to try to recoup some of the production costs before the goods become unsellable.

5. One-of-a-kind goods:- Handmade art or other rare goods cannot be easily reproduced, so the supply cannot expand even if the price rises.

6. Inelastic supply:-For many goods, including agricultural products, it is difficult to quickly adjust the supply even if the price rises. It can take months or even years for crops to reach maturity and become available to customers. For example, an apple farmer who adds trees to their orchard won't be able to harvest the fruit for several years.

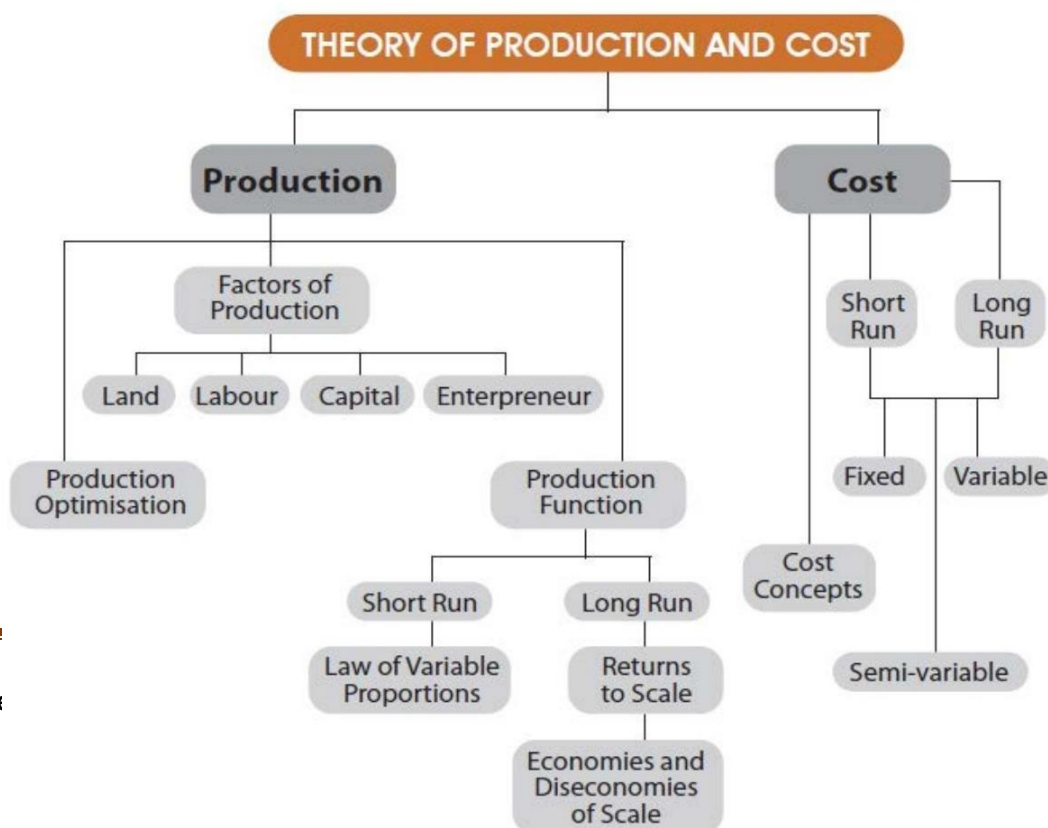
Ans.12 Explain the theory and factor of production and cost: -**MEANING OF PRODUCTION: -**

Production is a very important economic activity. As we are aware, the survival of any firm in a competitive market depends upon its ability to produce goods and services at a competitive cost. One of the principal concerns of business managers is the achievement of optimum efficiency in production by minimizing the cost of production. The performance of an economy is judged by the level of its production. The amount of goods and services an economy is able to produce determines the richness or poverty of that economy. In fact, the standard of living of people depends on the volume and variety of goods and services produced in a country. Thus, the U.S.A. is a rich country just because its level of production is high.

In common parlance, the term 'production' is used to indicate an activity of making something material. The growing of wheat, rice or any other agricultural crop by farmers and manufacturing of cement, radio-sets, wool, machinery or any other industrial product is often referred to as production.

What exactly do we mean by production in Economics?

In Economics the word 'production' is used in a wider sense to denote the process by which man utilizes resources such as men, material, capital, time etc, working upon them to transform them into commodities and services so as to make them satisfy human wants. In other words, production is any economic activity which converts inputs into outputs which are capable of satisfying human wants. Whether it is making of material goods or providing a service, it is included in production provided it satisfies the wants of some

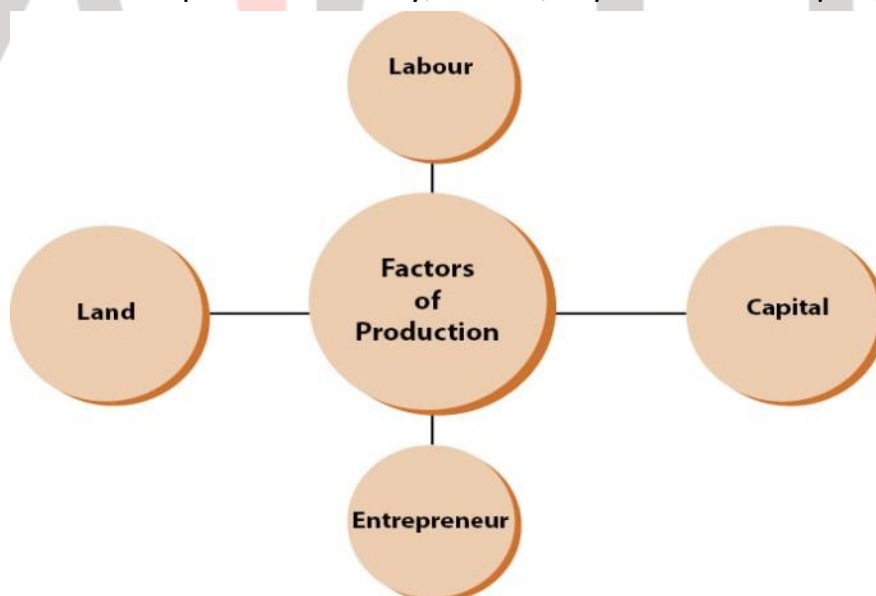


people. Therefore, in Economics, activities such as making of cloth by an industrial worker, the services of the retailer who delivers it to consumers, the work of doctors, lawyers, teachers, actors, dancers, etc. are production.

According to James Bates and J.R. Parkinson “Production is the organized activity of transforming resources into finished products in the form of goods and services; and the objective of production is to satisfy the demand of such transformed resources”.

FACTORS OF PRODUCTION: -

Factors of production refer to inputs. An input is a good or service which a firm buys for use in its production process. Production process requires a wide variety of inputs, depending on the nature of output. The process of producing goods in a modern economy is very complex. A good has to pass through many stages and many hands until it reaches the consumers' hands in a finished form. Land, labour, capital and entrepreneurial ability are the four factors or resources which make it possible to produce goods and services. Even a small piece of bread cannot be produced without the active participation of these factors of production. While land is a free gift of nature and refers to natural resources, the human endeavour is classified functionally and qualitatively into three main components namely, labour, capital and entrepreneurial skills.



- **Land** includes all those free natural resources whose supply for the economy as a whole is fixed.
- **Labour(Men)** is all human efforts of body or of mind undergone partly or

wholly with a view to secure an income apart from the pleasure derived directly from the work

- **Capital(Money)** is that part of wealth of an individual or community which is used for further production of wealth. Capital, a stock concept, refers to produced means of production and it comprises of manmade machines and materials which are used for further production.
- **Entrepreneur** is the person who organises business; initiates production, remunerates other factors of production, introduces innovations and bears the risk and uncertainties of business.

Factors of production can be divided into three categories –:

A. Input and output: -

Input means factors of production such as 5 M's Money(capital), Manpower (Labour), Methods, material & Machinery

Output means production (Commodity to be produced). In other words, an output is any commodity that comes out of production process.

B. Fixed factors and variable factors: -

FIXED FACTOR (FF)

- Fixed factors refer to those factors of production which cannot be changed during short run.
- These are used in a fixed quantity in the short run.
- These factors can be changed only in the long run.
- Example- land, plant and machinery, factory building etc.

VARIABLE FACTOR (VF)

- Variable factor refers to those factors of production which can be
-

changed during short period.

- The quantity of variable inputs varies according to the level of output.
- Example- labour, raw material etc.

C. Short run and long run: -

- ❖ **Short run:** Short run refers to a period of time in which a firm cannot change its fixed factors of production, only variable factors can be changed.
- ❖ **Long run:** Long run refers to a time period during which a firm can change all the factors of production. In the long run, all inputs are variable. Therefore the distinction between fixed factors and variable factors will disappear.

Production function: -

Production function is the technical relationship between inputs and output. Samuelson describes production function as the relationship between the maximum amount of output that can be produced and the input required to make that output. It is defined for a given state of technology.

1. Law of variable proportion

The law of variable proportion or the law of diminishing returns is relevant when some factors are kept fixed and others are varied. It is applicable to the short-run. There are three stages of the law of variable proportion – where:-

- law of increasing returns,
- law of diminishing returns and
- law of negative returns operates

2. The Law of returns to scale

The Law of returns to scale describes the relationship between inputs and

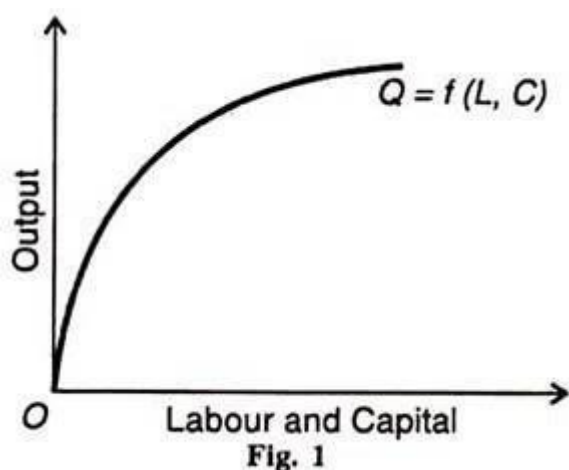
output in the long run when all inputs are changed in the same proportion. Returns to scale may be constant, increasing and decreasing.

- Constant returns to scale occur when the inputs increase by some proportion and the output also increases by the same proportion. It is also called linear homogeneous production function.
- Increasing returns to scale occur when the inputs increase by some proportion and the output increases more than proportionately.
- Decreasing returns to scale occur when the inputs increase by some proportion and the output increases less than proportionately.

3. Production optimization

Normally, a profit maximizing firm is interested to know what combination of factors of production (or inputs) would minimize its cost of production for a given output. This can be known by combining the firm's production and cost functions, namely isoquants and iso-cost lines respectively.

Production function can be stated as $Q_x = F(L, C)$



Suppose there are two-factor inputs: labour (L) and capital (K). Q_x is the quantity of output of commodity x, f is the function, L is the unit of labour and k is the unit capital. According to this, the amount of output depends on the labour

and capital inputs utilised in manufacturing.

Ans.13 What is the Features of Monopoly Market. Describe the difference between Perfect, Monopoly and Monopolistic Competition?

Monopoly: -

Pure Monopoly is the form of market organization in which there is a single seller of a commodity for which there are no close substitutes. Thus, it is at the opposite extreme from perfect competition monopoly may be the result of: (1) Increasing returns to scale; (2) Control over the supply of raw materials; (3) Patents; (4) Government Franchise.



Monopoly Examples



MONOPOLY (ECONOMICS)

Monopoly is a market structure in which a single seller or producer of a particular product or service dominates the industry.

OVERVIEW

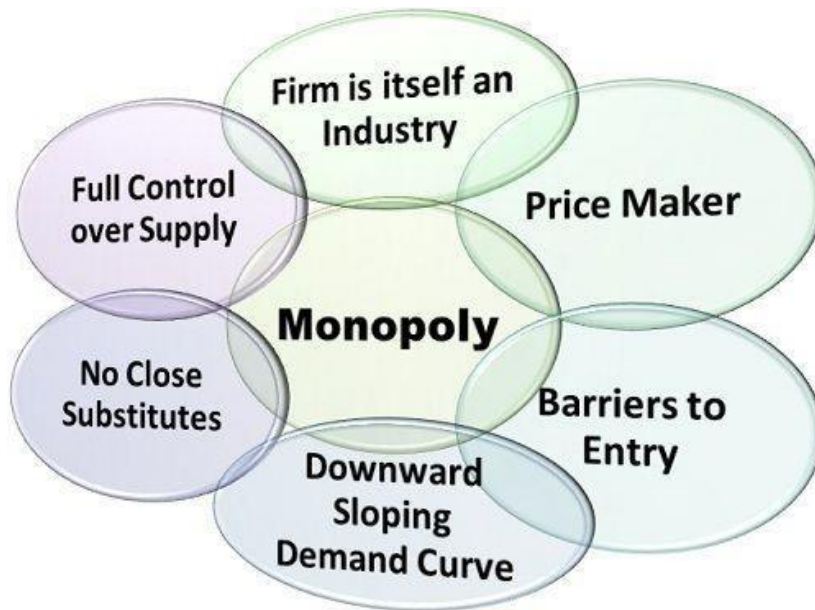
Monopoly is an economic term that refers to a market structure where one firm or producer has exclusive control over the production or sale of a good or service, resulting in no direct competition (Robinson, 1969). This gives the monopolist extreme power to set prices and manipulate market demand for that particular product.

EXAMPLE

Microsoft Corporation was once accused of using monopolistic tactics in the software industry. With almost all personal computers running on Windows in the 1990s, Microsoft had gained significant bargaining power over software developers and consumers, who had no other choice but to use Microsoft products. They were accused of unfairly raising prices and quashing competitors.

Features of Monopoly: -

1. **Single Seller:-**There is only one producer of a product. It may be due to some natural conditions prevailing in the market, or may be due to some legal restriction in the form of patents, copyright, sole dealership, state monopoly, etc. Since, there is only one seller any change in supply plans of that seller can have substantial influence over the market price. That is why a Monopolist is called a Price Maker.
2. **No Close substitute:-**The commodity sold by the Monopolist has no close substitute available for it. Therefore, if a consumer does not want the commodity at a particular price, he is likely to get available closely similar to what he is giving up. For example, there are chapters you have studied that the availability of substitute goods impact.
3. **Barriers to the entry of new firms:-**There are barriers to entry into industry for the new firms. It may be due to following reasons:
 - (i) Ownership of strategic raw material or exclusive knowledge of production
 - (ii) Patent Rights
 - (iii) Government Licensing
 - (iv) Natural Monopolies.
4. **Price Discrimination:-**Price Discrimination exists when the same product is sold at different price to different buyers. A monopolist practices price discrimination to maximize profits. For example Electricity Charges in Delhi are different for Domestic users and Commercial and Industrial users.
5. **Abnormal Profits in the Long run:-**Being the single seller, monopolists enjoy the benefit of higher profits in the long run.
6. **Limited Consumer Choice:-**As they are the single producer of the commodity, in the absence of any close substitute the choice for consumer is limited.
7. **Price in Excess of Marginal Cost:-**Monopolists fix the price of a commodity (per unit) higher than the cost of producing one additional unit as they have absolute control over Price Determination.



Let us compare the key characteristics of perfect competition and monopoly.

Basis	Perfect Competition	Monopoly
Number of Sellers	A large number of firms.	Single firm.
Barriers to Entry	Very low.	Very high.
Nature and availability of Substitute Products	Very good substitutes are readily available.	No good substitutes are available.
Firms compete through	Prices only.	Product features and quality, advertising, and marketing.
Pricing Power	Negligible. Dependent on demand and supply.	Significant. Companies can manipulate prices as they want.

The following table will help students grasp the most critical points of **Difference between monopoly and monopolistic competition.**



Monopoly	Monopolistic Competition
What does it mean?	
A monopoly is the type of imperfect competition where a seller or producer captures the majority of the market share due to the lack of substitutes or competitors.	A monopolistic competition is a type of imperfect competition where many sellers try to capture the market share by differentiating their products.
Number of players	
One	Many
Degree of competition	
No competition exists, as only one seller is present in the market.	A very high competition exists, as there are many sellers.
Barriers to entry	
High barriers to entry	Low barriers to entry
Demand curve	
Steep	Flat
Price control	
Due to steep demands and only one seller, the price is controlled by the seller.	Some level of price control is exercised by buyers, as many sellers are available in the market.

Advantages and Disadvantages of Monopolistic Competition: -

Monopolistic competition provides both benefits and pitfalls for companies and consumers.

Pros

- Few barriers to entry for new companies
- Variety of choices for consumers
- Company decision-making power for prices and marketing
- Consistent quality of product for consumers

Cons

- Many competitors limits access to economies of scale.
- Inefficient company spending on marketing, packaging and advertising
- Too many choices for consumers means extra research for consumers
- Misleading advertising or imperfect information for consumers.

Ans.14 From the following particulars, prepare a balance sheet of Mr. Somesh as at 31st March 2023: -

Capital ₹ 5,50,000
Drawings ₹10,000
Sundry Debtors ₹1,00,000
Sundry Creditors ₹80,000
Loan from Bank ₹20,000
Net profit ₹1,60,000
Closing Stock ₹50,000
Plant and Machinery ₹1,50,000
Building ₹1,20,000
Land ₹ 3,00,000
Goodwill ₹ 50,000
Furniture and Fixtures ₹30,000

Solution-:

Balance Sheet of Mr. Somesh

as at 31st March, 2021

Liabilities	₹	Assets	₹
Capital 5,50,000		Fixed / Non Current Assets:	
Add: Net Profit <u>1,60,000</u>		Goodwill	50,000
7,10,000		Building	1,20,000
		Land	3,00,000
Less: Drawings <u>10,000</u>	7,00,000	Plant & Machinery	1,50,000
Long-term Liabilities:		Furniture & Fixtures	30,000
Loan from bank	20,000		
Current liabilities:		Current Assets:	
Sundry Creditors	80,000	Closing stock	50,000
	<u>8,00,000</u>	Sundry Debtors	1,00,000
			<u>8,00,000</u>

Note: -A **Drawing** in accounting terms includes any money that is taken from the business account for personal use.

Ans: 15

Journalize the following transactions in the books of Laxman.

	₹
Business started with cash	80,000
Purchased furniture from Modern Furnishers for cash	10,000
Deposited in bank	30,000
Purchased goods for cash	5,000
Purchased goods on credit from Hari	4,500
Purchased goods from Suresh for cash	3,000
Purchased goods from Ajay	4,000
Sold goods for cash	7,000
Sold goods to Karim	6,500
Sold goods to Ganesh for cash	5,000
Sold goods to Bansi	3,500
Withdrew from bank for office use	5,000
Karim paid on account	5,000
Paid rent for residence	6,000
Paid to Ajay in settlement of his account	3,800

In the Book of Laxman- Journal of Laxman

Date	Particular	Dr. (Rs.)	Cr. (Rs.)
1	Cash a/c Dr. To Capital A/c	80,000	80,000
2	Furniture A/c Dr. To Cash A/c	10,000	10,000
3	Bank A/c Dr. To cash A/c	30,000	30,000

4	Purchase A/c	Dr.	5,000	
	To Cash A/c			5,000
5	Purchase A/c	Dr.	4,500	
	To Hari A/c			4,500
6	Purchase a/c	Dr.	3,000	
	To Cash A/c			3,000
7	Purchase A/c	Dr.	4,000	
	To Cash A/c			4,000
8	Cash a/c	Dr.	7,000	
	To Sales A/c			7,000
9	Cash a/c	Dr.	6500	
	To Sales A/c			6500
10	Cash a/c	Dr.	5,000	
	To Sales A/c			5,000
11	Cash a/c	Dr.	3500	
	To Sales A/c			3500
12	Cash a/c	Dr.	5,000	
	To Bank			5,000
13	Bank A/c	Dr.	5,000	
	To Karim A/c			5,000
14	Drawing A/c	Dr.	6000	

	To Cash A/c		6000
15	Ajay a/c Dr.	4,000	
	To Cash A/c		3,800
	To Discount A/c		200

Note :- 1. The golden rule for real accounts is: debit what comes in and credit what goes out. In this transaction, cash goes out and the loan is settled. Hence, in the journal entry, the Loan account will be debited and the Bank account will be credited.